AAE 590 LGM HW4

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1 The Banana Distribution

The std for the velocity noise was changed to 0.1.

1.1 Simulate the unicycle.

See Appendix for the code.

1.2 Plot the distribution.

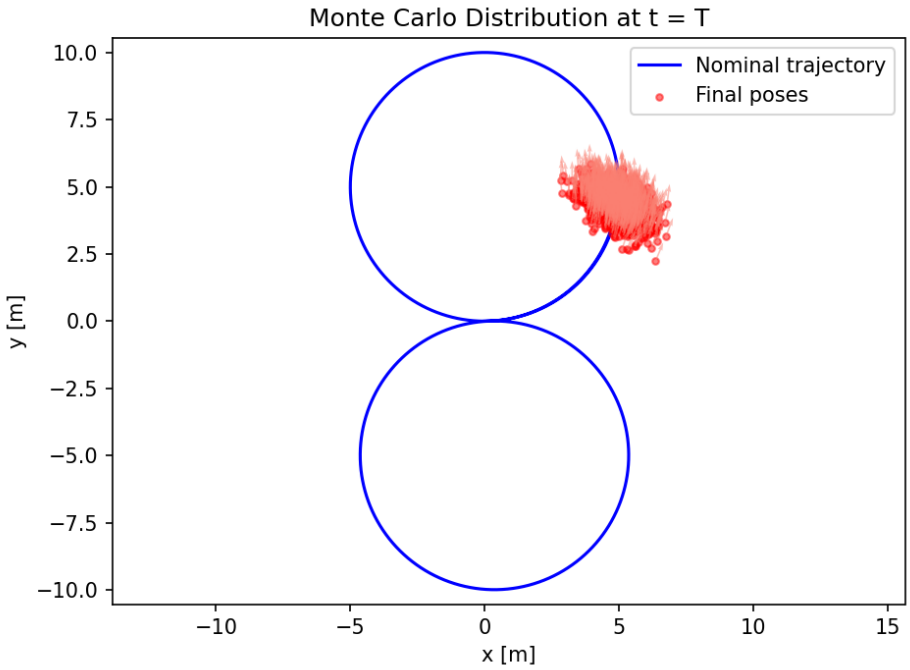


Figure 1: Nominal trajectory and Monte Carlo distribution.

1.3 Fit a Gaussian.

$$\bar{p} = \begin{bmatrix} 4.9350 \\ 4.2802 \end{bmatrix} \text{ m}$$

$$S = \begin{bmatrix} 2.1308 & 0.0429 \\ 0.0429 & 2.1889 \end{bmatrix} \text{ m}^2$$

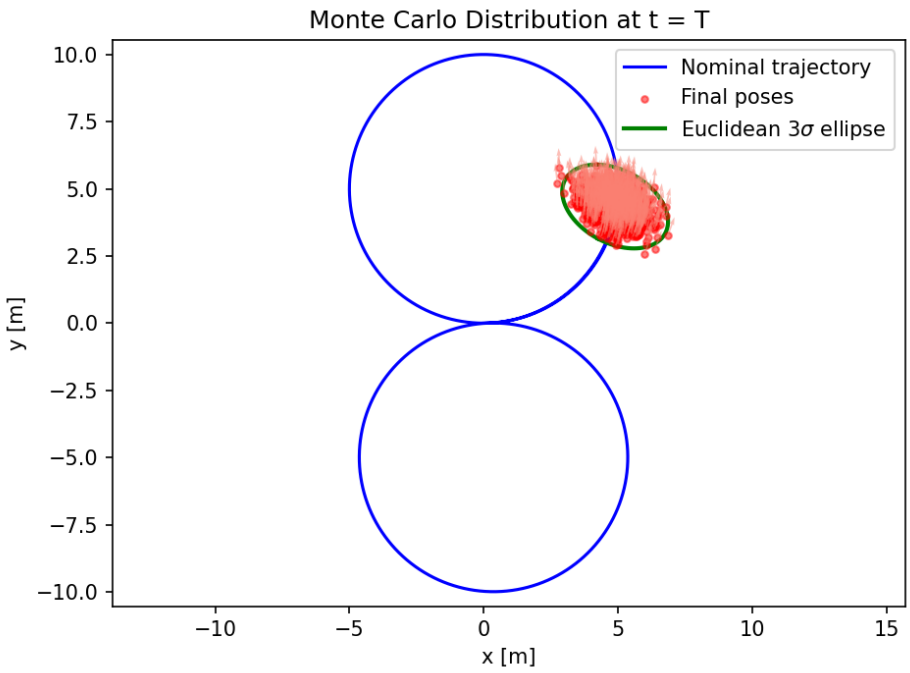


Figure 2: Gaussian Fit

The Gaussian fits the shape of the point cloud.

1.4 Characterize the distribution.

The overall shape looks like an ellipse with maybe slightly deformation. This is because we only have noise in the vehicle forward direction and combining it with the movement along the arc, where we have angular velocity and its uncertainty, it accumulates the error that produces a banana shape. The Gaussian ellipse cannot properly approximate it because it assumes symmetric distribution around mean which is not true in our case.

1.5 Intermediate snapshots.

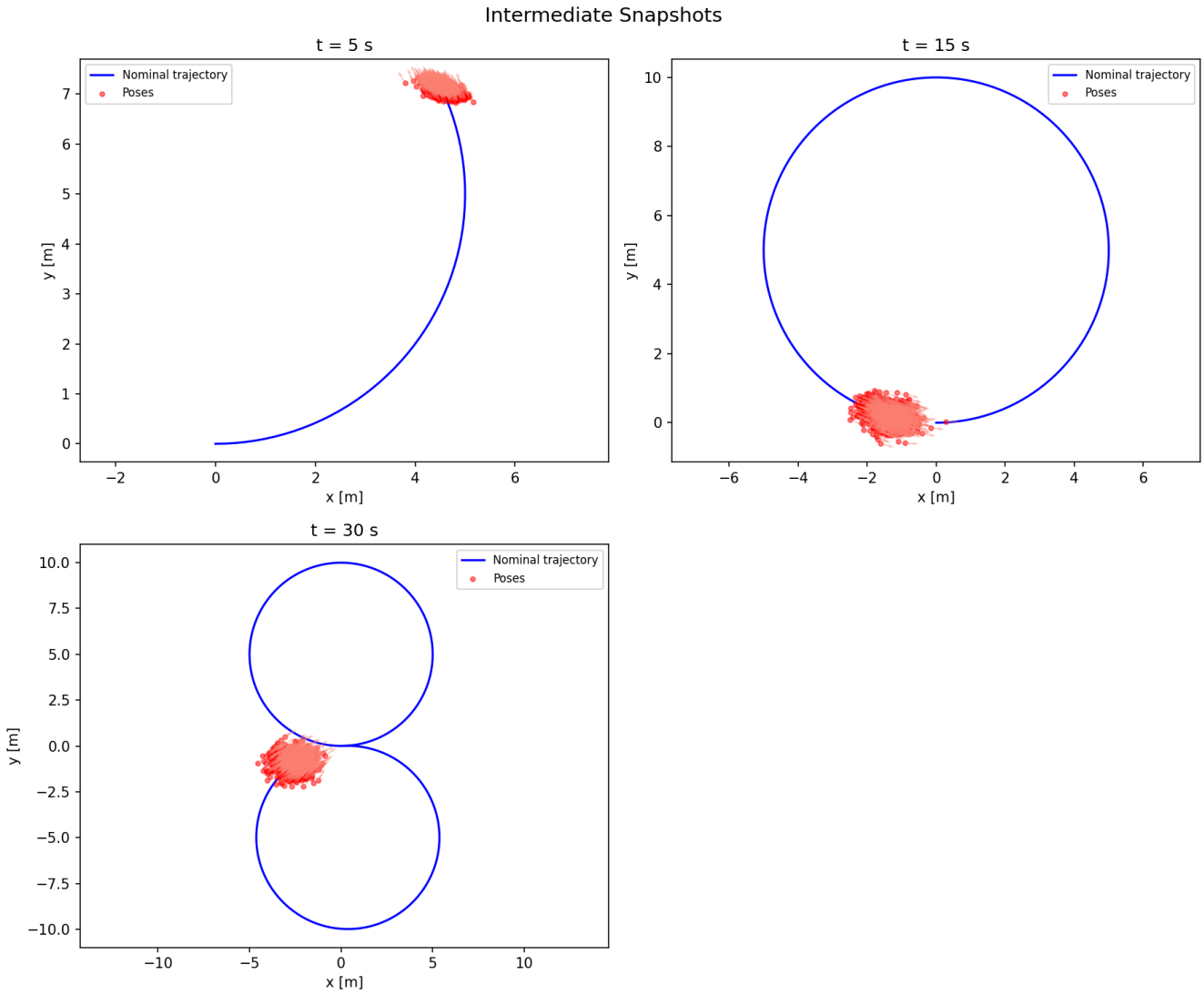


Figure 3: Snapshots

At $t = 5\ s$ we can see an oval shape and at other snapshots the distributions gets more dispersed.

1.6 Lie algebra Gaussian fit.

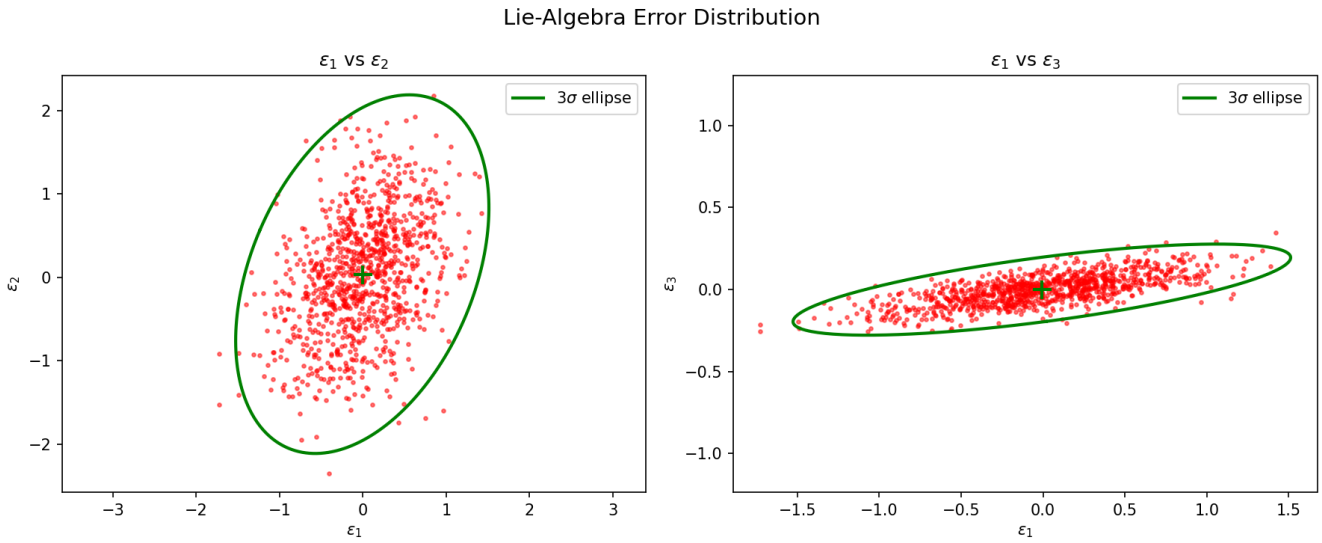


Figure 4: Gaussian Distribution In Lie algebra

$$\bar{\varepsilon} = \begin{bmatrix} -0.0338 \\ -0.0459 \\ -0.00085 \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} 2.0661 & -0.0299 & 0.0314 \\ -0.0299 & 2.3565 & 0.0427 \\ 0.0314 & 0.0427 & 0.00806 \end{bmatrix}$$

The clouds in both projections are approximately ellipsoidal.

1.7 Exponentiate the ellipsoid boundary.

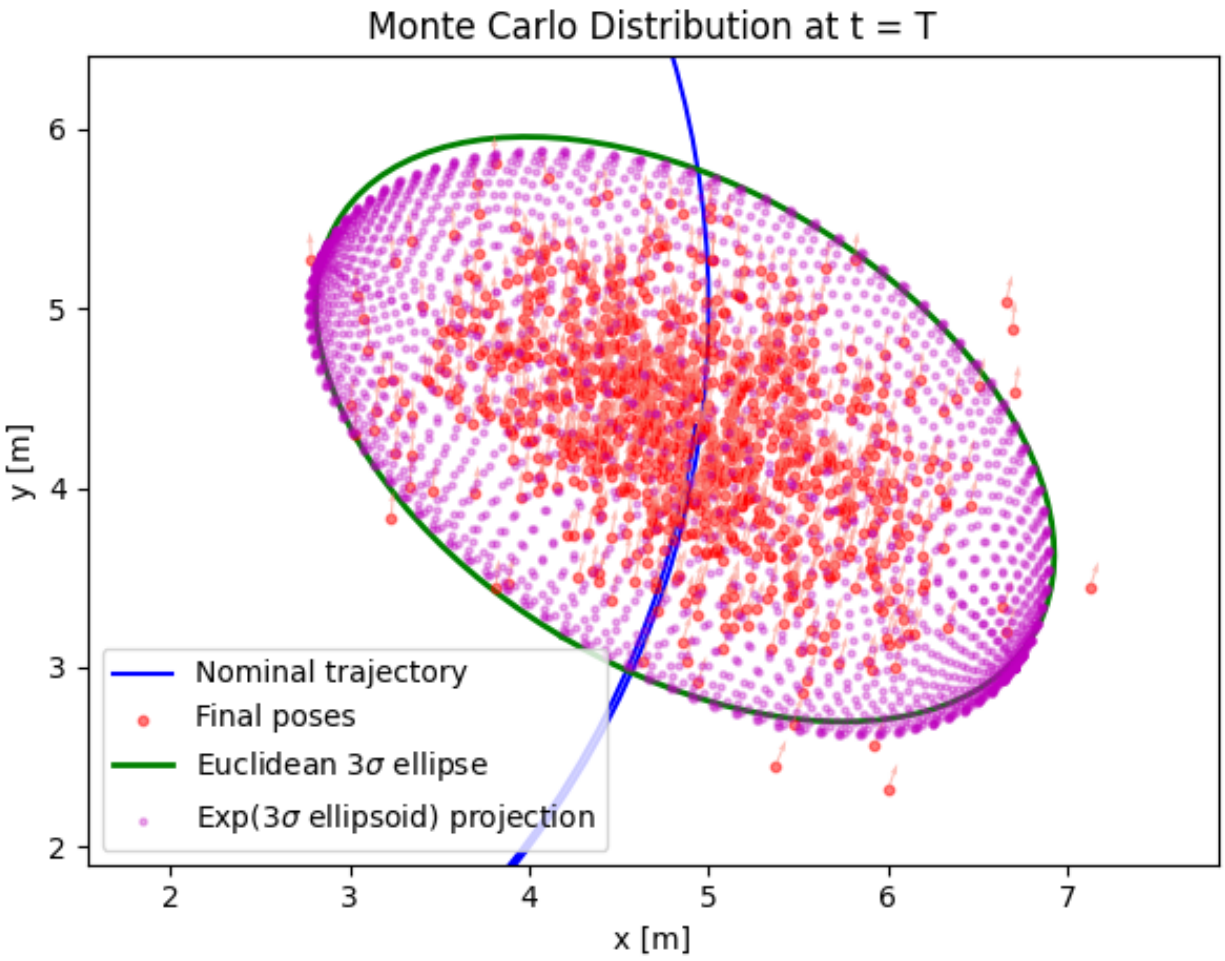


Figure 5: Lie algebra ellipsoid projection

We can see some deviation from an ideal ellipsoid which resembles the banana shape. A straight ellipsoid in the Lie algebra maps to a curved shape because the exponential map is non linear because the heading component couples into translational part.

2 World-Frame EKF

2.1 Prediction step.

$$F_k = \begin{bmatrix} 1 & 0 & -\Delta t\, v_{\text{nom}} \sin(\theta) \\ 0 & 1 & \Delta t\, v_{\text{nom}} \cos(\theta) \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} Q_k^{(\text{world})} &= \mathbb{E}\big[(L_k \mathbf{w})(L_k \mathbf{w})^\top\big] \\ &= \mathbb{E}\big[L_k\, \mathbf{w}\, \mathbf{w}^\top\, L_k^\top\big] \\ &= L_k\, \mathbb{E}\big[\mathbf{w}\, \mathbf{w}^\top\big]\; L_k^\top \\ &= L_k\, Q_{\text{body}}\, L_k^\top . \end{aligned}$$

where $L_k = \Delta t \begin{bmatrix} \cos\hat{\theta}_k & -\sin\hat{\theta}_k & 0 \\ \sin\hat{\theta}_k & \cos\hat{\theta}_k & 0 \\ 0 & 0 & 1 \end{bmatrix}$

But because we have to lateral slip, we have no noise there and the second column is zero.

$L_k = \Delta t \begin{bmatrix} \cos\hat{\theta}_k & 0 & 0 \\ \sin\hat{\theta}_k & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

The covariance rotates because the noise is attached to the body frame, but our filter works in the world frame, so we have to rotate the body-frame covariance into world coordinates. Since this rotation depends on the current heading $\hat{\theta}_k$, which changes every timestep, the world-frame covariance reorients accordingly.

2.2 Measurement update.

$H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

2.3 Implementation

See Appendix for the code.

2.4 Animation.

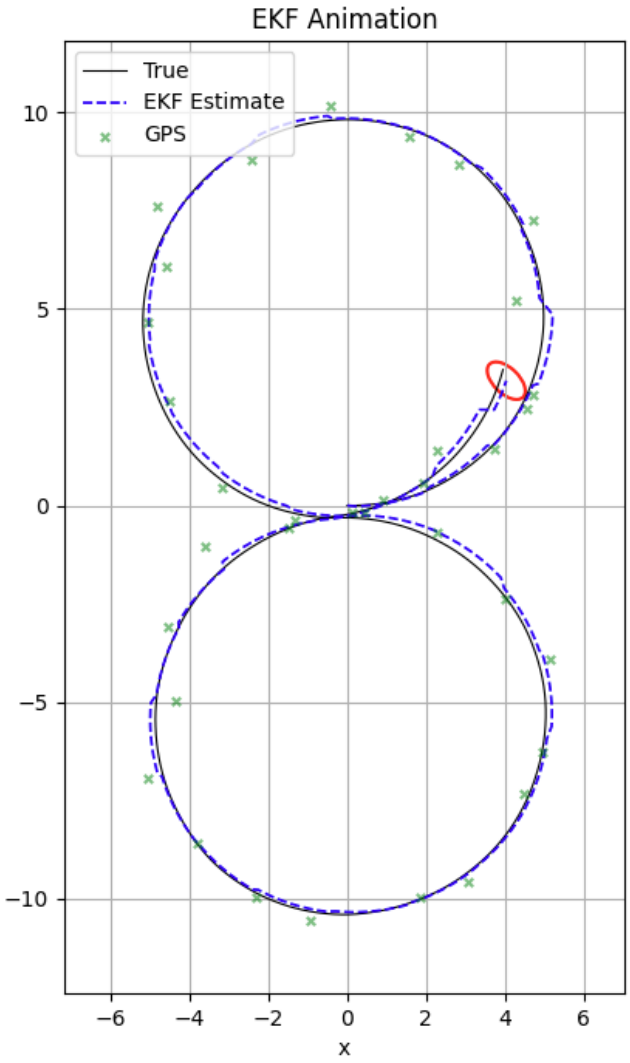


Figure 6: Unicycle with EKF

The long axis of the covariance ellipse is never aligned with the trajectory of the unicycle. It is close to be perpendicular to it. However if we increase the std of velocity to 1 m/s, the long axis is almost always aligned with the trajectory.

2.5 Monte Carlo evaluation.

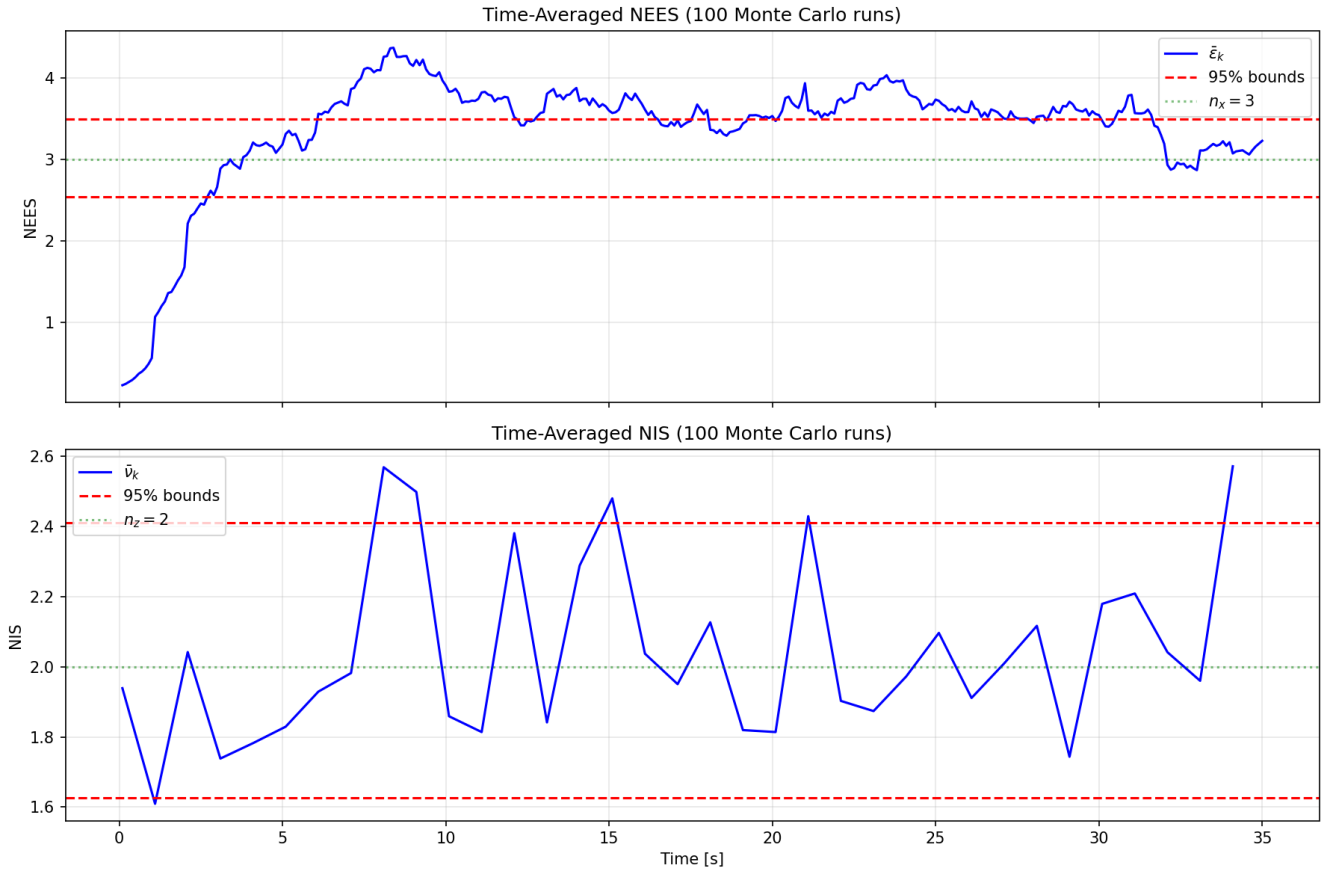


Figure 7: Time averages NEES and NIS

The NEES is consistently above 95% line which suggests that our filter is overconfident. It also has low values in the first 3 seconds, which indicates that our filter is under-confident with the initial value of P.

The NIS stays stable throughout the run and averages around 2, which is the measurement dimension. The advantage of this metric is that we can calculate it without the true state, which means we can monitor the filter's performance onboard the controller.

3 Error-State EKF in the Body Frame

3.1 Error dynamics.

1. Position differences (world frame)

True and reference dynamics for x:

$$x_{k+1} = x_k + (v_{\text{nom}} + w_v)\cos\theta_k\,\Delta t,$$
$$\hat{x}_{k+1} = \hat{x}_k + v_{\text{nom}}\cos\hat{\theta}_k\,\Delta t.$$

Subtracting,

$$x_{k+1} - \hat{x}_{k+1} = (x_k - \hat{x}_k) + v_{\text{nom}}\Delta t\big(\cos\theta_k - \cos\hat{\theta}_k\big) + w_v\Delta t\cos\theta_k.$$

Expanding $\cos\theta_k = \cos(\hat{\theta}_k + \delta\theta)$ to first order:

$$\cos\theta_k \approx \cos\hat{\theta}_k\underbrace{\cos\delta\theta}_{\approx 1} - \sin\hat{\theta}_k\underbrace{\sin\delta\theta}_{\approx\delta\theta} = \cos\hat{\theta}_k - \delta\theta\sin\hat{\theta}_k,$$

so $\cos\theta_k - \cos\hat{\theta}_k \approx -\delta\theta \sin\hat{\theta}_k$, giving

$$x_{k+1} - \hat{x}_{k+1} \approx (x_k - \hat{x}_k) - v_{\text{nom}}\Delta t\, \delta\theta \sin\hat{\theta}_k + w_v \Delta t \cos\hat{\theta}_k.$$

Analogously for y:

$$\begin{aligned} y_{k+1} &= y_k + (v_{\text{nom}} + w_v)\sin\theta_k\, \Delta t, \\ \hat{y}_{k+1} &= \hat{y}_k + v_{\text{nom}} \sin\hat{\theta}_k\, \Delta t, \\ \sin\theta_k &\approx \sin\hat{\theta}_k + \delta\theta\cos\hat{\theta}_k, \\ y_{k+1} - \hat{y}_{k+1} &\approx (y_k - \hat{y}_k) + v_{\text{nom}}\Delta t\, \delta\theta \cos\hat{\theta}_k + w_v \Delta t \sin\hat{\theta}_k. \end{aligned}$$

2.Expand trig at $\hat{\theta}_{k+1}$

Since $\hat{\theta}_{k+1} = \hat{\theta}_k + \omega_{\text{ref}}\Delta t$,

$$\begin{aligned} \cos\hat{\theta}_{k+1} &\approx \cos\hat{\theta}_k - \sin\hat{\theta}_k\, \omega_{\text{ref}}\Delta t, \\ \sin\hat{\theta}_{k+1} &\approx \sin\hat{\theta}_k + \cos\hat{\theta}_k\, \omega_{\text{ref}}\Delta t. \end{aligned}$$

3. Body-frame error δx_b^{k+1}

Let $C = \cos\hat{\theta}_k$, $S = \sin\hat{\theta}_k$ for brevity. Then

$$\delta x_b^{k+1} = \cos\hat{\theta}_{k+1}(x_{k+1} - \hat{x}_{k+1}) + \sin\hat{\theta}_{k+1}(y_{k+1} - \hat{y}_{k+1}).$$

Substituting everything and labeling terms:

$$\begin{aligned} \delta x_b^{k+1} = (C - S\omega_{\text{ref}}\Delta t)\big[&\underbrace{(x_k - \hat{x}_k)}_{(1)} - \underbrace{v_{\text{nom}}\Delta t\, \delta\theta\, S}_{(2)} + \underbrace{w_v \Delta t\, C}_{(3)}\big] \\ + (S + C\omega_{\text{ref}}\Delta t)\big[&\underbrace{(y_k - \hat{y}_k)}_{(4)} + \underbrace{v_{\text{nom}}\Delta t\, \delta\theta\, C}_{(5)} + \underbrace{w_v \Delta t\, S}_{(6)}\big]. \end{aligned}$$

Collect terms, dropping any product of two small quantities (two of $\delta\theta, \delta x_b, \delta y_b, w_v, w_\omega, \Delta t$ beyond first order):

$$\begin{aligned} C\cdot(1) + S\cdot(4) &= C(x_k - \hat{x}_k) + S(y_k - \hat{y}_k) = \delta x_b^k, \\ -S\omega_{\text{ref}}\Delta t\cdot(1) + C\omega_{\text{ref}}\Delta t\cdot(4) &= \omega_{\text{ref}}\Delta t\big[-S(x_k - \hat{x}_k) + C(y_k - \hat{y}_k)\big] = \omega_{\text{ref}}\Delta t\, \delta y_b^k, \\ C\cdot(2) + S\cdot(5) &= -v_{\text{nom}}\Delta t\, \delta\theta\, CS + v_{\text{nom}}\Delta t\, \delta\theta\, SC = 0, \\ C\cdot(3) + S\cdot(6) &= w_v\Delta t\, C^2 + w_v\Delta t\, S^2 = w_v\Delta t, \end{aligned}$$

with all remaining cross-terms $\mathcal{O}(\Delta t^2)$ or products of two small quantities. Therefore:

$$\boxed{\delta x_b^{k+1} \approx \delta x_b^k + \omega_{\text{ref}}\Delta t\, \delta y_b^k + w_v \Delta t.}$$

4. Body-frame error δy_b^{k+1}

$$\delta y_b^{k+1} = -\sin\hat{\theta}_{k+1}(x_{k+1} - \hat{x}_{k+1}) + \cos\hat{\theta}_{k+1}(y_{k+1} - \hat{y}_{k+1}).$$

Using the same expansion,

$$\begin{aligned} \delta y_b^{k+1} = &-(S + C\omega_{\text{ref}}\Delta t)\big[(x_k - \hat{x}_k) - v_{\text{nom}}\Delta t\, \delta\theta\, S + w_v\Delta t\, C\big] \\ &+ (C - S\omega_{\text{ref}}\Delta t)\big[(y_k - \hat{y}_k) + v_{\text{nom}}\Delta t\, \delta\theta\, C + w_v\Delta t\, S\big]. \end{aligned}$$

Collecting:

$$\begin{aligned} -S(x_k - \hat{x}_k) + C(y_k - \hat{y}_k) &= \delta y_b^k, \\ -C\omega_{\text{ref}}\Delta t(x_k - \hat{x}_k) - S\omega_{\text{ref}}\Delta t(y_k - \hat{y}_k) &= -\omega_{\text{ref}}\Delta t\, \delta x_b^k, \\ v_{\text{nom}}\Delta t\, \delta\theta(S^2 + C^2) &= v_{\text{nom}}\Delta t\, \delta\theta, \\ w_v\Delta t(-SC + CS) &= 0. \end{aligned}$$

Therefore:

$$\boxed{\delta y_b^{k+1} \approx -\omega_{\text{ref}}\Delta t\, \delta x_b^k + \delta y_b^k + v_{\text{nom}}\Delta t\, \delta\theta.}$$

5. Heading error

$$\theta_{k+1} - \hat{\theta}_{k+1} = \delta\theta + w_\omega\Delta t \quad \Longrightarrow \quad \boxed{\delta\theta^{k+1} = \delta\theta + w_\omega\Delta t.}$$

6. Assembled linearization

$$\varepsilon_{k+1} \;\approx\; A_k\,\varepsilon_k + G\,n_k,$$

with

$$A_k = \begin{bmatrix} 1 & \omega_{\text{ref}}\Delta t & 0 \\ -\omega_{\text{ref}}\Delta t & 1 & v_{\text{nom}}\Delta t \\ 0 & 0 & 1 \end{bmatrix}, \qquad G = \begin{bmatrix} \Delta t & 0 & 0 \\ 0 & \Delta t & 0 \\ 0 & 0 & \Delta t \end{bmatrix} = \Delta t\, I_3, \qquad n_k = \begin{bmatrix} w_v \\ 0 \\ w_\omega \end{bmatrix}.$$

A_k depends only on the commanded inputs $(v_{\text{nom}}, \omega_{\text{ref}})$ and not on the reference pose $(\hat{x}_k, \hat{y}_k, \hat{\theta}_k)$
Since we don't have to rotate out covariance matrix anymore, the covariance prediction is then

$$P_{k+1|k} = A_k P_{k|k} A_k^\top + G\,Q_{\text{body}}\,G^\top.$$

3.2 Observation: process noise is natural.

In error-state Kalman Filter the state (which is error) is defined in body frame. The covariance matrix represents noise for the body frame vectors, so no rotation is required since it is already in the body frame.

3.3 Measurement update.

$z_k = [x_k^w, y_k^w]^T + v_k$
$[x_k^w, y_k^w]^T = [\hat{x}_k^w, \hat{y}_k^w]^T + [\delta x_k^w, \delta y_k^w]^T$
$[\delta x_k^w, \delta y_k^w]^T = \hat{R}_k[\delta x_k^b, \delta y_k^b]^T$
$[x_k^w, y_k^w]^T = [\hat{x}_k^w, \hat{y}_k^w]^T + \hat{R}_k[\delta x_k^b, \delta y_k^b]^T$
$z_k \approx \hat{p}_k + \hat{R}_k[\delta x_k^b, \delta y_k^b]^T + v_k$

So, $H_k = [\hat{R}_k\ \mathbf{0_2}]$

3.4 Implementation.

See Appendix for the code.

3.5 Animation and comparison.

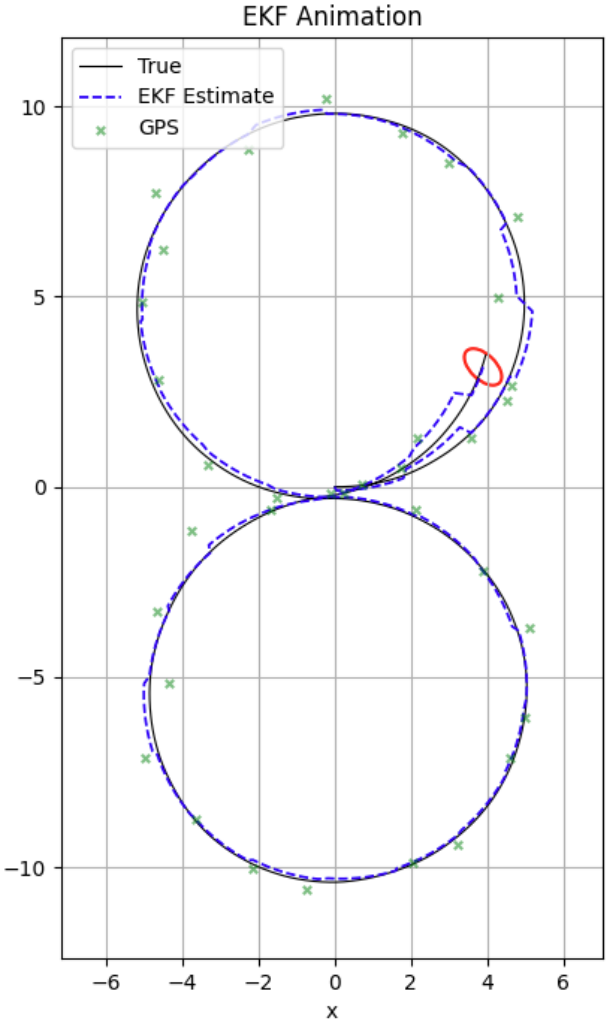


Figure 8: Unicycle with ES-EKF

We can see similar behavior as for the world frame EKF. For small std v, the long axis of the ellipse is never aligned with the trajectory, but for large std v the long axis is aligned with the trajectory. This might be a better match for banana distribution because the error is defined in body frame.

3.6 NEES and NIS comparison.

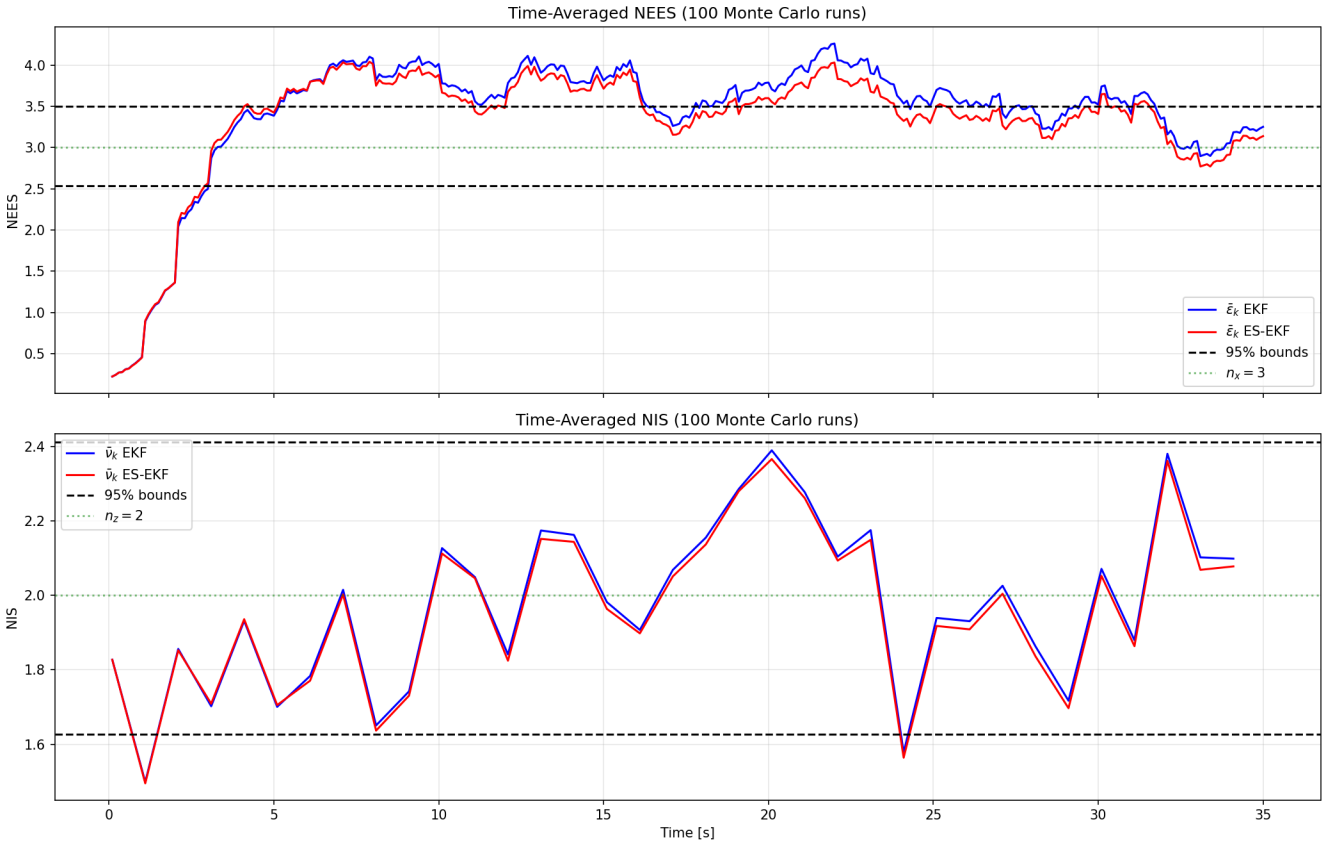


Figure 9: EKF vs ES-EKF

For NEES, the ES-EKF stays closer to the state dimension of 3. For NIS, ES-EKF is slightly better, but both filters stay around 2.

4 Left-Invariant IEKF

4.1 Left-invariant error dynamics via Lie algebra.

$\eta_k \approx \exp(\varepsilon_k^\wedge), \qquad \varepsilon_k = \log(\eta_k)^\vee$

$\hat{X}_{k+1} = \hat{X}_k \exp(\hat{\xi}^\wedge \Delta t) \quad \hat{X}_{k+1}^{-1} = \exp(\hat{\xi}^\wedge \Delta t)^{-1} \hat{X}_k^{-1} \quad X_{k+1} = X_k \exp\big((\hat{\xi} + n_k)^\wedge \Delta t\big)$

$\varepsilon_{k+1} =$

$$
\begin{aligned}
&= \log(\eta_{k+1}) \\
&= \log\big(\hat{X}_{k+1}^{-1} X_{k+1}\big) \\
&= \log\Big(\exp(\hat{\xi}^\wedge \Delta t)^{-1} \hat{X}_k^{-1} X_k \exp\big((\hat{\xi} + n_k)^\wedge \Delta t\big)\Big) \\
&= \log\Big(\exp(-\hat{\xi}^\wedge \Delta t)\, \hat{X}_k^{-1} X_k \, \exp(\hat{\xi}^\wedge \Delta t)\, \exp(n_k^\wedge \Delta t)\Big) \\
&= \log\Big(\exp(-\hat{\xi}^\wedge \Delta t)\, \exp(\varepsilon_k^\wedge)\, \exp(\hat{\xi}^\wedge \Delta t)\Big) + \log\big(\exp(n_k^\wedge \Delta t)\big) \\
&= \log\Big(\exp\big(\underbrace{\mathrm{Ad}_{\exp(-\hat{\xi}^\wedge \Delta t)}\, \varepsilon_k^\wedge}_{\mathrm{Ad}_{\exp(-\hat{\xi}^\wedge \Delta t)}} \big)\Big) + n_k \Delta t \\
&= \mathrm{Ad}_{\exp(-\hat{\xi}^\wedge \Delta t)}\, \varepsilon_k + n_k \Delta t \\
&= A_k\, \varepsilon_k + n_k \Delta t
\end{aligned}
$$

$A_k = \mathrm{Ad}_{\exp(-\hat{\xi}^\wedge \Delta t)}.$

$A_c = \left. \dfrac{d}{d\Delta t}\, \mathrm{Ad}_{\exp(-\hat{\xi}^\wedge \Delta t)} \right|_{\Delta t = 0}$

$\mathrm{Ad}_{\exp(-\hat{\xi}^\wedge \Delta t)}(\varepsilon^\wedge) = \exp(-\hat{\xi}^\wedge \Delta t)\, \varepsilon^\wedge\, \exp(\hat{\xi}^\wedge \Delta t)$

Differentiating in Δt by the product rule:

$\dfrac{d}{d\Delta t}\Big[\exp(-\hat{\xi}^\wedge \Delta t)\, \varepsilon^\wedge\, \exp(\hat{\xi}^\wedge \Delta t)\Big] = -\hat{\xi}^\wedge \exp(-\hat{\xi}^\wedge \Delta t)\, \varepsilon^\wedge\, \exp(\hat{\xi}^\wedge \Delta t) + \exp(-\hat{\xi}^\wedge \Delta t)\, \varepsilon^\wedge\, \hat{\xi}^\wedge \exp(\hat{\xi}^\wedge \Delta t).$

Evaluating at $\Delta t = 0$ (so $\exp(0) = I$)

$\left. \dfrac{d}{d\Delta t} \mathrm{Ad}_{\exp(-\hat{\xi}^\wedge \Delta t)}(\varepsilon^\wedge) \right|_{\Delta t = 0} = -\hat{\xi}^\wedge \varepsilon^\wedge + \varepsilon^\wedge \hat{\xi}^\wedge = -[\hat{\xi}^\wedge, \varepsilon^\wedge] = -\mathrm{ad}_{\hat{\xi}}(\varepsilon^\wedge)$

Hence $A_c = -\mathrm{ad}_{\hat{\xi}}$

Computing $\mathrm{ad}_{\hat{\xi}}$ by the commutator formula.

With $\hat{\xi} = [\hat{v}, 0, \hat{\omega}]^\top$ and $\varepsilon = [\delta x_b, \delta y_b, \delta\theta]^\top$, the hat maps are

$$
\hat{\xi}^\wedge = \begin{bmatrix} 0 & -\hat{\omega} & \hat{v} \\ \hat{\omega} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \varepsilon^\wedge = \begin{bmatrix} 0 & -\delta\theta & \delta x_b \\ \delta\theta & 0 & \delta y_b \\ 0 & 0 & 0 \end{bmatrix}.
$$

$$
\hat{\xi}^\wedge \varepsilon^\wedge = \begin{bmatrix} -\hat{\omega}\,\delta\theta & 0 & -\hat{\omega}\,\delta y_b \\ 0 & -\hat{\omega}\,\delta\theta & \hat{\omega}\,\delta x_b \\ 0 & 0 & 0 \end{bmatrix}, \qquad \varepsilon^\wedge \hat{\xi}^\wedge = \begin{bmatrix} -\hat{\omega}\,\delta\theta & 0 & 0 \\ 0 & -\hat{\omega}\,\delta\theta & \delta\theta\,\hat{v} \\ 0 & 0 & 0 \end{bmatrix}.
$$

$$
[\hat{\xi}^\wedge, \varepsilon^\wedge] = \begin{bmatrix} 0 & 0 & -\hat{\omega}\,\delta y_b \\ 0 & 0 & \hat{\omega}\,\delta x_b - \hat{v}\,\delta\theta \\ 0 & 0 & 0 \end{bmatrix},
$$

$$\mathrm{ad}_{\hat{\xi}}\,\varepsilon = \begin{bmatrix} -\hat{\omega}\,\delta y_b \\ \hat{\omega}\,\delta x_b - \hat{v}\,\delta\theta \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & -\hat{\omega} & 0 \\ \hat{\omega} & 0 & -\hat{v} \\ 0 & 0 & 0 \end{bmatrix} \varepsilon.$$

Therefore

$$A_c = -\mathrm{ad}_{\hat{\xi}} = \begin{bmatrix} 0 & \hat{\omega} & 0 \\ -\hat{\omega} & 0 & \hat{v} \\ 0 & 0 & 0 \end{bmatrix}.$$

As we can see, both A_k and A_c depend only on the error and not the state.

$$P_{k+1|k} = A_k P_{k|k} A_k^\top + G\,Q_{\text{body}}\,G^\top.$$

4.2 Measurement Jacobian.

The measurement function h extracts the translation part of $X \in SE(2)$. Write $\hat{X}_k$ and the perturbation in block form:

$$\hat{X}_k = \begin{bmatrix} \hat{R}_k & \hat{p}_k \\ 0 & 1 \end{bmatrix}, \qquad \mathrm{Exp}(\varepsilon^\wedge) = \begin{bmatrix} R(\delta\theta) & V(\delta\theta)\begin{bmatrix}\delta x_b \\ \delta y_b\end{bmatrix} \\ 0 \qquad\qquad & 1 \end{bmatrix},$$

where $V(\delta\theta)$ is the SE(2) left Jacobian. To first order in ε,

$$R(\delta\theta) \approx I + \delta\theta\,J, \qquad V(\delta\theta) \approx I, \qquad J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix},$$

so

$$\mathrm{Exp}(\varepsilon^\wedge) \approx \begin{bmatrix} I + \delta\theta\,J & \begin{bmatrix}\delta x_b \\ \delta y_b\end{bmatrix} \\ 0 \qquad & 1 \end{bmatrix}$$

Multiplying,

$$\hat{X}_k\,\mathrm{Exp}(\varepsilon^\wedge) = \begin{bmatrix} \hat{R}_k(I + \delta\theta\,J) & \hat{p}_k + \hat{R}_k\begin{bmatrix}\delta x_b \\ \delta y_b\end{bmatrix} \\ 0 \qquad\qquad & 1 \end{bmatrix}$$

The measurement h picks off the translation block:

$$h\Big(\hat{X}_k\,\mathrm{Exp}(\varepsilon^\wedge)\Big) = \hat{p}_k + \hat{R}_k\begin{bmatrix}\delta x_b \\ \delta y_b\end{bmatrix}$$

$$z_k = \hat{p}_k + \hat{R}_k\begin{bmatrix}\delta x_b \\ \delta y_b\end{bmatrix} + v_k$$

So

$$\boxed{H_k = \begin{bmatrix}\hat{R}_k & \mathbf{0}_2\end{bmatrix} \in \mathbb{R}^{2\times 3}.}$$

4.3 Implementation.

See Appendix for the code.

4.4 Full comparison.

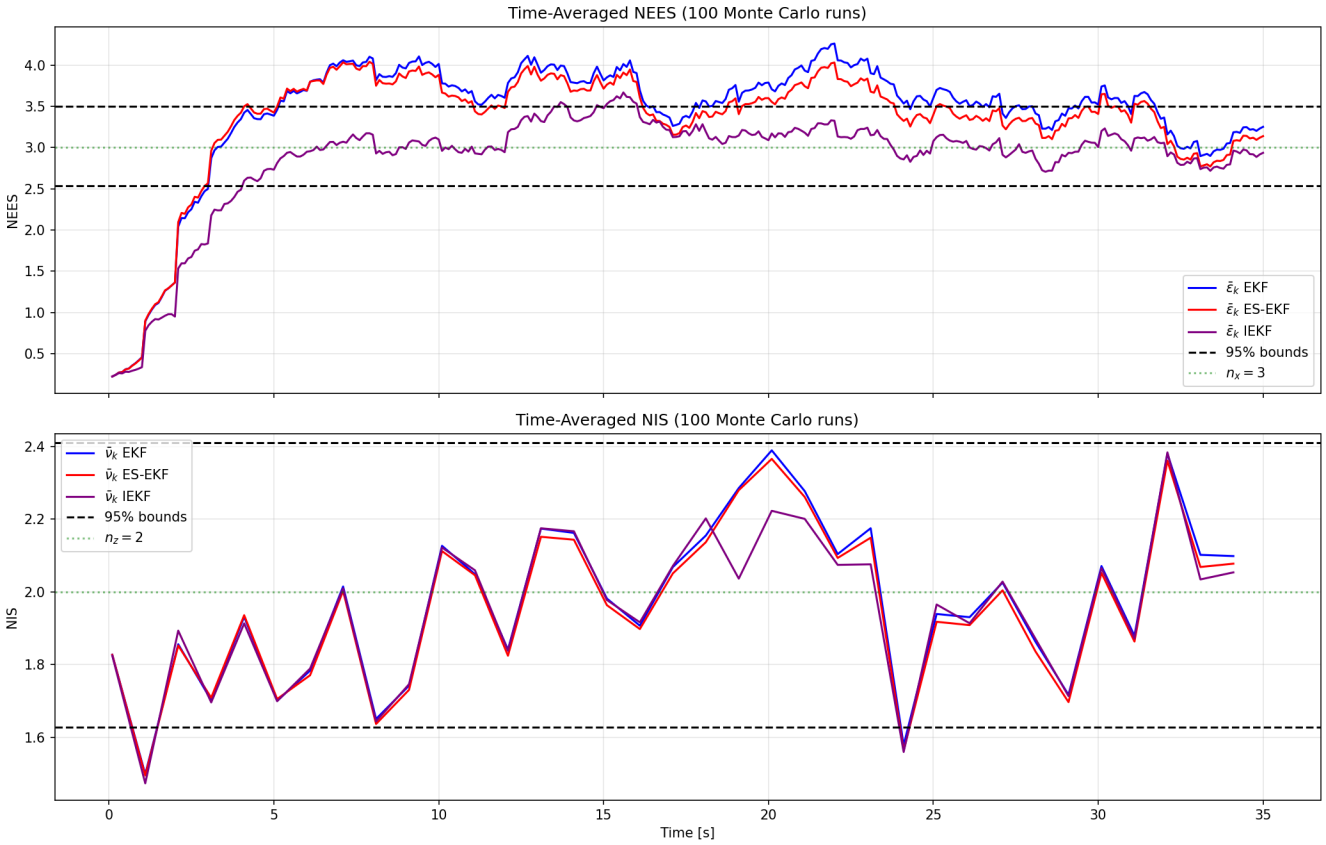


Figure 10: Filter comparison

Filter	Time-averaged NEES	Time-averaged NIS
World-frame EKF	3.4338	1.9709
Error-state EKF	3.3302	1.9567
Left-invariant IEKF	2.8742	1.9509

Table 1: Filter consistency comparison over 100 Monte Carlo runs.

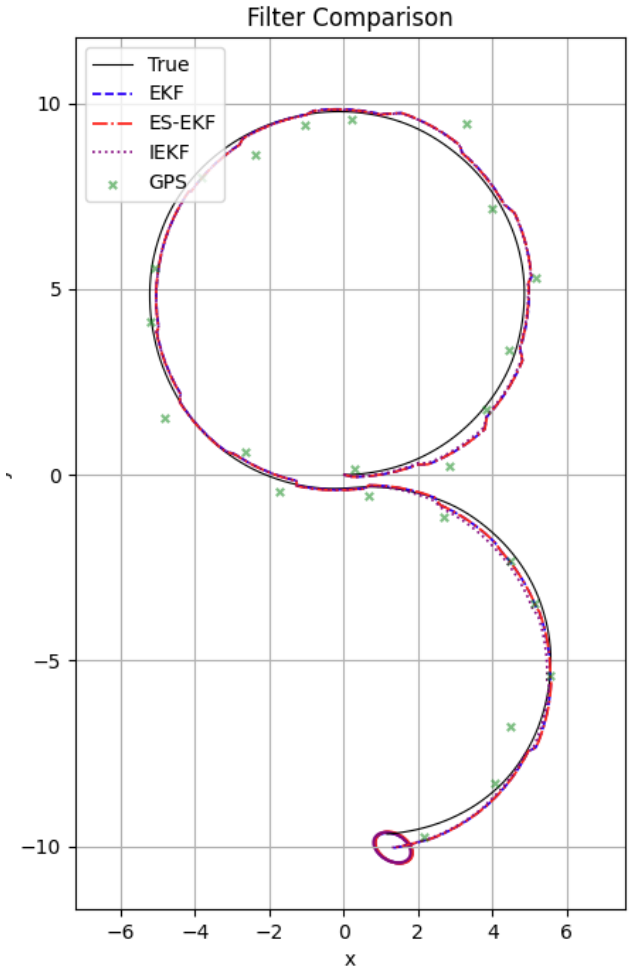


Figure 11: Covariance ellipses

4.5 Discussion.

The regular EKF has a covariance in the world frame so to plot it, we need to convert it to body frame which is corrupted by uncertainty in heading. The Error State EKF has covariance in the body frame already, so it is better than EKF. The IEKF also has the covariance ellipsoid described in body frame, so the ellipsoid should be similar to Error State EKF. However, the fact that in IEKF the error lives in lie algebra, might make covariance ellipsoid to be closer to true banana distribution than other ellipsoids.

The banana shape exists because rotating and translating do not commute on SE(2). forward-velocity noise that accumulates while the vehicle turns produces a curved uncertainty cloud that cannot be captured by a straight ellipse in world coordinates. The world frame EKF rotates body noise into world coordinates at every step using current heading, so any heading error contaminates how forward velocity uncertainty is accumulated which produces nonlinearity. Working in the body frame keeps the process noise in its natural frame and delays the nonlinear coupling until the measurement update, so the filter no longer linearizes through the very nonlinearity that produces the banana shape.

The state independence property is the one that would be good to preserve. This makes the filter to be dependent only on control inputs and not the state, which eliminates the corruption of the estimation from the uncertainties in the state. This gives a consistency and the global convergence guarantees for the estimated state.

5 Unicycle Simulation in Modelica with Rumoca

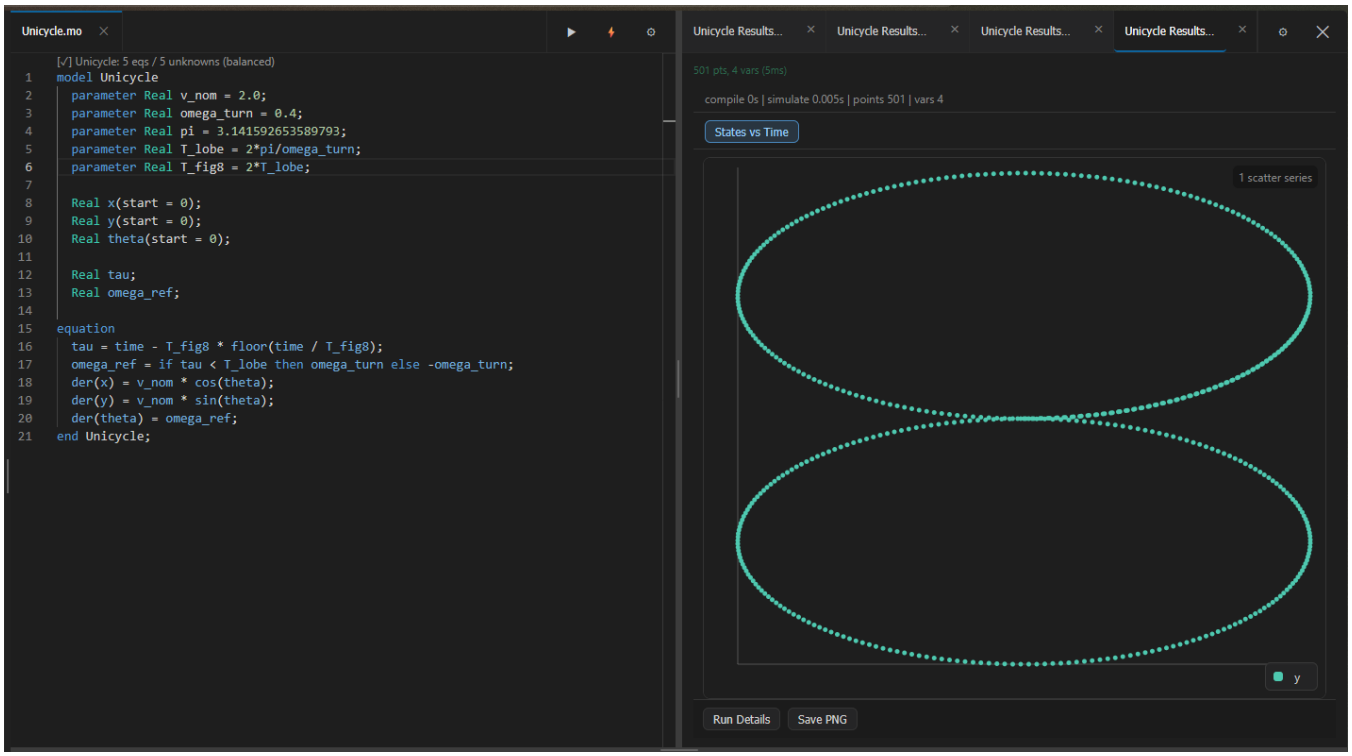


Figure 12: Modelica model

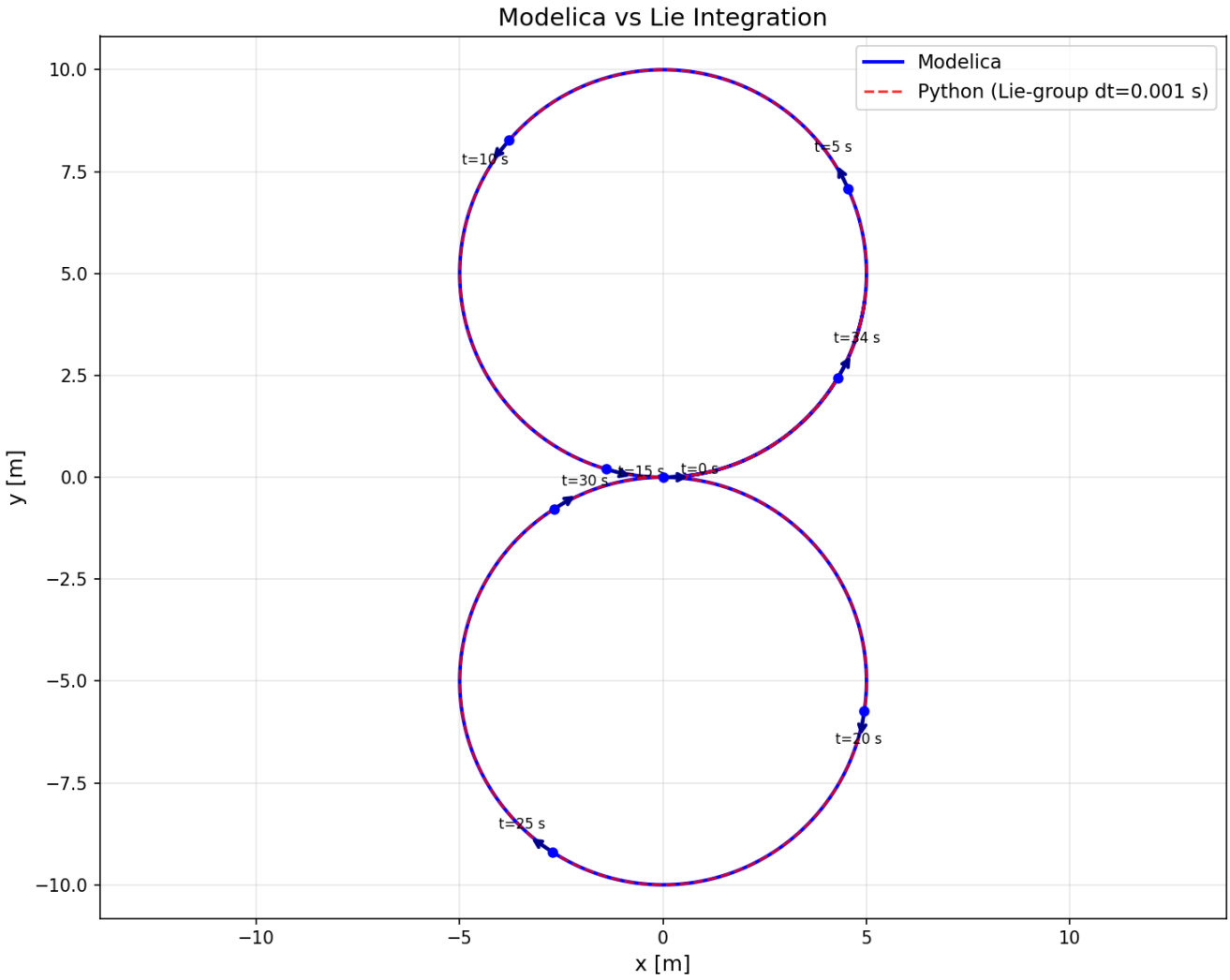


Figure 13: Unicycle Trajectory

The Modelica and Lie group trajectories are basically the same. However, on the lower lobe of the figure we see a difference. This is due to the time step of the Python method: the switch between negative and positive angular velocity does not happen instantaneously. In this case, the max difference is 7.361898e-01 m. If we decrease the time step to 0.001 s, the trajectories match almost exactly and the max difference drops to 2.946335e-04 m.

6 Appendix

Listing 1: Problem 1 code

```
import numpy as np
import matplotlib.pyplot as plt
import os, sys
sys.path.append(os.path.abspath(os.path.join(os.path.dirname(__file__), "..")))
from se2 import se2_exp, se2_log

pi = np.pi

v_nom = 2
```

```
w_turn = 0.4
r_fig = 5
sigma_v = 0.05
sigma_w = 0.5

T_lobe = 2 * pi / w_turn
T_fig8 = 10 * pi
T = 35

dt = 0.1

# Nominal trajectory
X0 = np.eye(3)
n_steps = int(T / dt)
X_ref = np.zeros((n_steps + 1, 3, 3))
X_ref[0] = X0
for i in range(n_steps):
    t = i * dt
    if t % T_fig8 < T_lobe:
        w_ref = w_turn
    else:
        w_ref = -w_turn
    xi = np.array([v_nom, 0, w_ref]) * dt
    X_ref[i + 1] = X_ref[i] @ se2_exp(xi)

## Monte Carlo simulation
N = 1000
snap_times = [5, 15, 30]
snap_steps = [int(ts / dt) for ts in snap_times]
X_snaps = {ts: np.zeros((N, 3, 3)) for ts in snap_times}
X_finals = np.zeros((N, 3, 3))

for j in range(N):
    w_v = np.random.normal(0, sigma_v, n_steps)
    w_w = np.random.normal(0, sigma_w, n_steps)
    X = X0.copy()
    for i in range(n_steps):
        t = i * dt
        if t % T_fig8 < T_lobe:
            w_ref = w_turn
        else:
            w_ref = -w_turn
        xi = np.array([v_nom + w_v[i], 0, w_ref + w_w[i]]) * dt
        X = X @ se2_exp(xi)
        for ts, ns in zip(snap_times, snap_steps):
            if i + 1 == ns:
                X_snaps[ts][j] = X
    X_finals[j] = X
    if j % 100 == 0:
        print(f"Iteration {j}/{N} completed")

# Extract final positions and headings
x_final = X_finals[:, 0, 2]
y_final = X_finals[:, 1, 2]
theta_final = np.arctan2(X_finals[:, 1, 0], X_finals[:, 0, 0])

# Statistics
positions = np.vstack([x_final, y_final])
p_bar = np.mean(positions, axis=1)
```

```
S = np.cov(positions)
print(f"Sample mean p_bar = [{p_bar[0]:.4f}, {p_bar[1]:.4f}]")
print(f"Sample covariance S =\n{S}")

# Covariance ellipse
t_ell = np.linspace(0, 2 * np.pi, 100)
circle = np.array([np.cos(t_ell), np.sin(t_ell)])
eigvals, eigvecs = np.linalg.eigh(S)
L = eigvecs @ np.diag(3 * np.sqrt(eigvals))
ellipse_pts = L @ circle + p_bar[:, None]

# Log-map errors: eps_i = Log(X_nom^{-1} @ X_i)^vee      R^3
eps = np.array([se2_log(np.linalg.inv(X_ref[-1]) @ X_finals[i]) for i in range(N)])
eps_bar = np.mean(eps, axis=0)
Sigma = np.cov(eps.T)
print(f"Lie-algebra sample mean eps_bar = {eps_bar}")
print(f"Lie-algebra sample covariance Sigma =\n{Sigma}")

# 3   ellipsoid boundary in Lie algebra, projected to (x, y) via Exp map
phi_grid = np.linspace(0, np.pi, 50)
psi_grid = np.linspace(0, 2 * np.pi, 50)
PHI, PSI = np.meshgrid(phi_grid, psi_grid)
sphere = np.array([np.sin(PHI) * np.cos(PSI),
                   np.sin(PHI) * np.sin(PSI),
                   np.cos(PHI)])  # (3, 100, 50)

eigvals_e, eigvecs_e = np.linalg.eigh(Sigma)
VD_half = eigvecs_e @ np.diag(3 * np.sqrt(eigvals_e))  # V @ (3*D^{1/2})

X_nom = X_ref[-1]
ell_x = np.zeros_like(PHI)
ell_y = np.zeros_like(PHI)
for ii in range(PHI.shape[0]):
    for jj in range(PHI.shape[1]):
        eps_pt = eps_bar + VD_half @ sphere[:, ii, jj]
        X_pt = X_nom @ se2_exp(eps_pt)
        ell_x[ii, jj] = X_pt[0, 2]
        ell_y[ii, jj] = X_pt[1, 2]

# --- Figure 1: final distribution at t = T ---
fig1, ax1 = plt.subplots()
ax1.plot(X_ref[:, 0, 2], X_ref[:, 1, 2], "b-", label="Nominal trajectory")
ax1.scatter(x_final, y_final, s=10, c="r", zorder=3, alpha=0.5, label="Final poses")
ax1.quiver(x_final, y_final, np.cos(theta_final), np.sin(theta_final),
           scale=40, width=0.002, color="salmon", zorder=4, alpha=0.5)
ax1.plot(ellipse_pts[0], ellipse_pts[1], "g-", lw=2, label=r"Euclidean $3\sigma$ ellipse"
    )
ax1.scatter(ell_x, ell_y, s=5, c="m", alpha=0.3, zorder=5,
            label=r"Exp($3\sigma$ ellipsoid) projection")
ax1.set_xlabel("x [m]")
ax1.set_ylabel("y [m]")
ax1.axis("equal")
ax1.legend()
ax1.set_title("Monte Carlo Distribution at t = T")
fig1.tight_layout()
fig1.savefig(os.path.join(os.path.dirname(__file__), "banana_final.png"), dpi=150)

# --- Figure 2: intermediate snapshots at t = 5, 15, 30 ---
fig2, axes = plt.subplots(2, 2, figsize=(12, 10))
```

```
axes[1, 1].set_visible(False)
plot_axes = [axes[0, 0], axes[0, 1], axes[1, 0]]
for ax, ts in zip(plot_axes, snap_times):
    Xs = X_snaps[ts]
    xs = Xs[:, 0, 2]
    ys = Xs[:, 1, 2]
    ths = np.arctan2(Xs[:, 1, 0], Xs[:, 0, 0])
    ref_idx = int(ts / dt)
    ax.plot(X_ref[:ref_idx + 1, 0, 2], X_ref[:ref_idx + 1, 1, 2], "b-",
            label="Nominal trajectory")
    ax.scatter(xs, ys, s=10, c="r", zorder=3, alpha=0.5, label="Poses")
    ax.quiver(xs, ys, np.cos(ths), np.sin(ths),
              scale=40, width=0.002, color="salmon", zorder=4, alpha=0.5)
    ax.set_xlabel("x [m]")
    ax.set_ylabel("y [m]")
    ax.axis("equal")
    ax.legend(fontsize=8)
    ax.set_title(f"t = {ts} s")
fig2.suptitle("Intermediate Snapshots", fontsize=14)
fig2.tight_layout()
fig2.savefig(os.path.join(os.path.dirname(__file__), "banana_snapshots.png"), dpi=150)

# --- Figure 3: Lie-algebra error projections (eps1-eps2 and eps1-eps3) ---
t_ell = np.linspace(0, 2 * np.pi, 200)
circle = np.array([np.cos(t_ell), np.sin(t_ell)])
labels = [r"$\varepsilon_1$", r"$\varepsilon_2$", r"$\varepsilon_3$"]
proj_pairs = [(0, 1), (0, 2)]

fig3, axes3 = plt.subplots(1, 2, figsize=(12, 5))
for ax, (i, j) in zip(axes3, proj_pairs):
    ax.scatter(eps[:, i], eps[:, j], s=5, c="r", alpha=0.5)
    sub_mean = eps_bar[[i, j]]
    sub_cov = Sigma[np.ix_([i, j], [i, j])]
    eigvals_s, eigvecs_s = np.linalg.eigh(sub_cov)
    L_s = eigvecs_s @ np.diag(3 * np.sqrt(eigvals_s))
    ell = L_s @ circle + sub_mean[:, None]
    ax.plot(ell[0], ell[1], "g-", lw=2, label=r"$3\sigma$ ellipse")
    ax.plot(sub_mean[0], sub_mean[1], "g+", ms=12, mew=2)
    ax.set_xlabel(labels[i])
    ax.set_ylabel(labels[j])
    ax.axis("equal")
    ax.legend()
    ax.set_title(f"{labels[i]} vs {labels[j]}")
fig3.suptitle("Lie-Algebra Error Distribution", fontsize=14)
fig3.tight_layout()
fig3.savefig(os.path.join(os.path.dirname(__file__), "banana_lie_errors.png"), dpi=150)

plt.show()
```

Listing 2: Unicycle modelica code

```
model Unicycle
  parameter Real v_nom = 2.0;
  parameter Real omega_turn = 0.4;
  parameter Real pi = 3.141592653589793;
  parameter Real T_lobe = 2*pi/omega_turn;
  parameter Real T_fig8 = 2*T_lobe;

  Real x(start = 0);
```

```
  Real y(start = 0);
  Real theta(start = 0);

  Real tau;
  Real omega_ref;

equation
  tau = time - T_fig8 * floor(time / T_fig8);
  omega_ref = if tau < T_lobe then omega_turn else -omega_turn;
  der(x) = v_nom * cos(theta);
  der(y) = v_nom * sin(theta);
  der(theta) = omega_ref;
end Unicycle;
```

Listing 3: Rumoca generated code

```

"""
Generated by Rumoca.

This template renders the current DAE context into a SymPy-friendly symbolic
model. It keeps Rumoca's continuous equations in residual form and can solve
for explicit state derivatives and algebraics when possible.
"""

import sympy as sp


def der(symbol):
    if isinstance(symbol, sp.Indexed):
        return sp.IndexedBase(f"{symbol.base}_dot")[symbol.indices]
    if isinstance(symbol, sp.IndexedBase):
        return sp.IndexedBase(f"{symbol}_dot")
    return sp.Symbol(f"{symbol}_dot")


def pre(symbol):
    if isinstance(symbol, sp.Indexed):
        return sp.IndexedBase(f"pre_{symbol.base}")[symbol.indices]
    if isinstance(symbol, sp.IndexedBase):
        return sp.IndexedBase(f"pre_{symbol}")
    return sp.Symbol(f"pre_{symbol}")


def if_else(condition, then_expr, else_expr):
    return sp.Piecewise((then_expr, condition), (else_expr, True))


def _vector(entries):
    return (
        sp.Matrix([symbol for _, symbol, _ in entries]),
        {name: start for name, _, start in entries},
        {name: index for index, (name, _, _) in enumerate(entries)},
    )


def _column(entries):
    return sp.Matrix(entries) if entries else sp.Matrix([])

```

```
44
45  def zeros(*dims):
46      if len(dims) == 1:
47          return sp.Matrix.zeros(int(dims[0]), 1)
48      return sp.Matrix.zeros(*(int(dim) for dim in dims))
49
50
51  def ones(*dims):
52      if len(dims) == 1:
53          return sp.Matrix.ones(int(dims[0]), 1)
54      return sp.Matrix.ones(*(int(dim) for dim in dims))
55
56
57  def sum(values):
58      if isinstance(values, sp.MatrixBase):
59          return sp.Add(*list(values))
60      if isinstance(values, (list, tuple)):
61          return sp.Add(*values)
62      return values
63
64
65  abs = sp.Abs
66  min = sp.Min
67  max = sp.Max
68  sign = sp.sign
69  sqrt = sp.sqrt
70  sin = sp.sin
71  cos = sp.cos
72  tan = sp.tan
73  asin = sp.asin
74  acos = sp.acos
75  atan = sp.atan
76  atan2 = sp.atan2
77  sinh = sp.sinh
78  cosh = sp.cosh
79  tanh = sp.tanh
80  exp = sp.exp
81  log = sp.log
82  log10 = lambda value: sp.log(value, 10)
83  floor = sp.floor
84  ceil = sp.ceiling
85  transpose = lambda value: sp.Matrix(value).T
86  identity = sp.eye
87  cross = lambda lhs, rhs: sp.Matrix(lhs).cross(sp.Matrix(rhs))
88  div = lambda lhs, rhs: sp.floor(lhs / rhs)
89  mod = sp.Mod
90  rem = sp.Mod
91
92
93  class Model:
94      """Flattened symbolic DAE model."""
95
96      def __init__(self):
97          self.time = sp.Symbol("time")
98          time = self.time
99
100         # States, algebraics, inputs, outputs, parameters, constants, discretes.
101
102
```

```
        theta = sp.Symbol("theta")


        x = sp.Symbol("x")


        y = sp.Symbol("y")


        self.x, self.x_start, self.x_index = _vector([
            ("theta", theta, 0),
            ("x", x, 0),
            ("y", y, 0)
        ])



        omega_ref = sp.Symbol("omega_ref")


        tau = sp.Symbol("tau")


        self.y, self.y_start, self.y_index = _vector([
            ("omega_ref", omega_ref, 0),
            ("tau", tau, 0)
        ])


        self.u, self.u_start, self.u_index = _vector([])


        self.w, self.w_start, self.w_index = _vector([])



        T_fig8 = sp.Symbol("T_fig8")


        T_lobe = sp.Symbol("T_lobe")


        omega_turn = sp.Symbol("omega_turn")


        pi = sp.Symbol("pi")


        v_nom = sp.Symbol("v_nom")


        self.p, self.p_start, self.p_index = _vector([
            ("T_fig8", T_fig8, (2 * T_lobe)),
            ("T_lobe", T_lobe, ((2 * pi) / omega_turn)),
            ("omega_turn", omega_turn, 0.4),
            ("pi", pi, 3.141592653589793),
            ("v_nom", v_nom, 2)
        ])

```

```
        self.constants, self.constants_start, self.constants_index = _vector([])


        self.z, self.z_start, self.z_index = _vector([])


        self.m, self.m_start, self.m_index = _vector([])


        self.x_dot_alias, self.x_dot_alias_start, self.x_dot_alias_index = _vector([])

        self.f_x = _column([
            (tau - (time - (T_fig8 * floor((time / T_fig8))))),
            (omega_ref - if_else((tau < T_lobe), omega_turn, (-omega_turn))),
            (der(x) - (v_nom * cos(theta))),
            (der(y) - (v_nom * sin(theta))),
            (der(theta) - omega_ref)
        ])

        self.initial_residuals = _column([])

        self.relations = _column([
            (tau < T_lobe)
        ])

        self.explicit_targets = [

            tau,


            omega_ref,


            der(x),


            der(y),


            der(theta),

        ]
        self.explicit_solution = None
        self.explicit_rhs = sp.Matrix([])

    def solve_explicit(self):
        """Best-effort symbolic solve for explicit derivatives and algebraics."""
        if len(self.f_x) == 0 or len(self.explicit_targets) == 0:
            self.explicit_solution = {}
            self.explicit_rhs = sp.Matrix([])
            return self.explicit_solution

        solutions = sp.solve(
            list(self.f_x),
            self.explicit_targets,
            dict=True,
            simplify=False,
        )
```

```
        self.explicit_solution = solutions[0] if solutions else {}
        self.explicit_rhs = sp.Matrix(
            [
                self.explicit_solution.get(symbol, symbol)
                for symbol in self.explicit_targets
            ]
        )
        return self.explicit_solution

    def summary(self):
        return {
            "states": [str(symbol) for symbol in self.x],
            "algebraics": [str(symbol) for symbol in self.y],
            "inputs": [str(symbol) for symbol in self.u],
            "parameters": [str(symbol) for symbol in self.p],
            "continuous_residual_count": len(self.f_x),
            "initial_residual_count": len(self.initial_residuals),
            "explicit_target_count": len(self.explicit_targets),
        }


if __name__ == "__main__":
    model = Model()
    print(model.summary())
    if len(model.f_x) > 0 and len(model.explicit_targets) > 0:
        print(model.solve_explicit())
```

Listing 4: Unicycle sim code

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp
from scipy.interpolate import interp1d
import sympy as sp
import sys, os

HERE = os.path.dirname(os.path.abspath(__file__))
sys.path.insert(0, os.path.join(HERE, ".."))
from se2 import se2_exp

# ============================================================
# Part (b): Build ODE from rumoca-generated symbolic model
# ============================================================
sys.path.insert(0, HERE)
from rumoca import Model, der

model = Model()
print("rumoca model summary:", model.summary())

targets = model.explicit_targets
residuals = list(model.f_x)
solution = {}
for target, residual in zip(targets, residuals):
    eq = residual.subs(solution)
    sol_list = sp.solve(eq, target)
    solution[target] = sol_list[0]

omega_turn_val = float(model.p_start["omega_turn"])
pi_val         = float(model.p_start["pi"])
```

```
v_nom_val      = float(model.p_start["v_nom"])
T_lobe_val     = 2 * pi_val / omega_turn_val
T_fig8_val     = 2 * T_lobe_val

param_subs = {sym: val for sym, val in zip(
    model.p, [T_fig8_val, T_lobe_val, omega_turn_val, pi_val, v_nom_val]
)}

theta_sym, x_sym, y_sym = model.x
dtheta_expr = solution[der(theta_sym)].subs(param_subs)
dx_expr     = solution[der(x_sym)].subs(param_subs)
dy_expr     = solution[der(y_sym)].subs(param_subs)

_f = sp.lambdify(
    (model.time, theta_sym),
    [dtheta_expr, dx_expr, dy_expr],
    modules="numpy",
)

def ode_rhs(t, state):
    theta = state[0]
    d = _f(t, theta)
    return [float(d[0]), float(d[1]), float(d[2])]

T = 35.0
x0 = [float(model.x_start[s]) for s in ["theta", "x", "y"]]
t_eval = np.linspace(0, T, 3501)

print(f"\nIntegrating ODE (RK45, rtol=1e-10) for T = {T} s ...")
sol = solve_ivp(
    ode_rhs, [0, T], x0,
    method="RK45",
    t_eval=t_eval,
    rtol=1e-10, atol=1e-12,
    max_step=0.01,
)
assert sol.success, f"solve_ivp failed: {sol.message}"
print(f"  nfev = {sol.nfev}")

t_mod     = sol.t
theta_mod = sol.y[0]
x_mod     = sol.y[1]
y_mod     = sol.y[2]

dt_py   = 0.1
n_steps = int(T / dt_py)
X_ref   = np.zeros((n_steps + 1, 3, 3))
X_ref[0] = np.eye(3)

for i in range(n_steps):
    t = i * dt_py
    tau = t % T_fig8_val
    w_ref = omega_turn_val if tau < T_lobe_val else -omega_turn_val
    xi = np.array([v_nom_val, 0, w_ref]) * dt_py
    X_ref[i + 1] = X_ref[i] @ se2_exp(xi)

t_py     = np.arange(n_steps + 1) * dt_py
x_py     = X_ref[:, 0, 2]
y_py     = X_ref[:, 1, 2]
```

```
theta_py = np.arctan2(X_ref[:, 1, 0], X_ref[:, 0, 0])

x_mod_at_py = interp1d(t_mod, x_mod, kind="cubic")(t_py)
y_mod_at_py = interp1d(t_mod, y_mod, kind="cubic")(t_py)

pos_err     = np.sqrt((x_mod_at_py - x_py)**2 + (y_mod_at_py - y_py)**2)
max_pos_err = np.max(pos_err)
idx_max     = np.argmax(pos_err)

print(f"  max_k ||p_Modelica - p_Python|| = {max_pos_err:.6e} m")
print(f"  at t = {t_py[idx_max]:.1f} s")


fig, ax = plt.subplots(figsize=(10, 8))

ax.plot(x_mod, y_mod, "b-", lw=2, label="Modelica")
ax.plot(x_py, y_py, "r--", lw=1.5, alpha=0.8,
        label=f"Python (Lie-group dt={dt_py} s)")

arrow_times = [0, 5, 10, 15, 20, 25, 30, 34]
for ta in arrow_times:
    idx = np.argmin(np.abs(t_mod - ta))
    cx, cy = x_mod[idx], y_mod[idx]
    dx = np.cos(theta_mod[idx])
    dy = np.sin(theta_mod[idx])
    ax.annotate(
        "", xy=(cx + 0.7 * dx, cy + 0.7 * dy), xytext=(cx, cy),
        arrowprops=dict(arrowstyle="->", color="darkblue", lw=2),
    )
    ax.plot(cx, cy, "bo", ms=5, zorder=5)
    ax.text(cx + 0.9 * dx, cy + 0.9 * dy, f"t={ta} s",
            fontsize=8, ha="center", va="bottom")

ax.set_xlabel("x [m]", fontsize=12)
ax.set_ylabel("y [m]", fontsize=12)
ax.set_title("Modelica vs Lie Integration",
             fontsize=14)
ax.legend(fontsize=11, loc="best")
ax.axis("equal")
ax.grid(True, alpha=0.3)
fig.tight_layout()
out_path = os.path.join(HERE, "unicycle_trajectory.png")
fig.savefig(out_path, dpi=150)
print(f"\nFigure saved to {out_path}")
plt.show()
```

In [16]:
```
import sympy
import numpy as np
import casadi as ca
import matplotlib.pyplot as plt
from scipy.stats import chi2
from matplotlib.patches import Ellipse
from matplotlib.animation import FuncAnimation
from IPython.display import HTML
import sys, os
sys.path.append(os.path.abspath('..'))
from se2 import se2_exp, se2_Ad
```

In [17]:
```
def ES_EKF():
    # Initialize variables
    x_nom           = ca.SX.sym("x_nom", 3)
    P         = ca.SX.sym("P", 3, 3)
    v         = ca.SX.sym("v")
    w         = ca.SX.sym("w")
    dt        = ca.SX.sym("dt")
    sigma_v   = ca.SX.sym("sigma_v")
    sigma_w   = ca.SX.sym("sigma_w")
    sigma_gps = ca.SX.sym("sigma_gps")
    z_gps     = ca.SX.sym("z_gps", 2)
    gps_flag  = ca.SX.sym("gps_flag")

    A_error = ca.vertcat(
    ca.horzcat( 1,      w*dt,  0     ),
    ca.horzcat(-w*dt,   1,     v*dt  ),
    ca.horzcat( 0,      0,     1     )
    )
    H = ca.vertcat(
    ca.horzcat(ca.cos(x_nom[2]), -ca.sin(x_nom[2]), 0),
    ca.horzcat(ca.sin(x_nom[2]),  ca.cos(x_nom[2]), 0)
    )
    Q_body = ca.diag(ca.vertcat(sigma_v**2, 0, sigma_w**2))
    Q_gps = ca.diag(ca.vertcat(sigma_gps**2, sigma_gps**2))

    # Propagate state
    x_prop = ca.vertcat(x_nom[0] + v*ca.cos(x_nom[2])*dt,
                        x_nom[1] + v*ca.sin(x_nom[2])*dt,
                        x_nom[2] + w*dt)

    # Prediction
    P_pred = A_error @ P @ A_error.T + Q_body*dt**2

    p = ca.vertcat(x_prop[0], x_prop[1])

    # Update
    y = z_gps - p
    S = H @ P_pred @ H.T + Q_gps
    K = P_pred @ H.T @ ca.inv(S)
    P_meas = (ca.SX.eye(3) - K @ H) @ P_pred @ (ca.SX.eye(3) - K @ H).T + K @ Q_gps
    de = K @ y
```

```
    x_meas = ca.vertcat(x_prop[0] + ca.cos(x_prop[2])*de[0] - ca.sin(x_prop[2])*de[
                        x_prop[1] + ca.sin(x_prop[2])*de[0] + ca.cos(x_prop[2])*de[
                        x_prop[2] + de[2])

    x_new = ca.if_else(gps_flag, x_meas, x_prop)
    P_new = ca.if_else(gps_flag, P_meas, P_pred)
    nu = ca.if_else(gps_flag, y.T @ ca.inv(S) @ y, np.nan)


    f_filter = ca.Function(
        "error_state_filter",
        [
            x_nom,
            P,
            v,
            w,
            dt,
            sigma_v,
            sigma_w,
            sigma_gps,
            z_gps,
            gps_flag
        ],
        [x_new, P_new, nu],
        [
            "x_nom",
            "P",
            "v",
            "w",
            "dt",
            "sigma_v",
            "sigma_w",
            "sigma_gps",
            "z_gps",
            "gps_flag"
        ],
        ["x_new", "P_new", "nu"],
    )
    return {"error_state_filter": f_filter}
```

In [18]:
```
def IEKF(X, P, v, w, dt, sigma_v, sigma_w, sigma_gps, z_gps, gps_flag):

    xi = np.array([v, 0, w])
    theta = np.arctan2(X[1, 0], X[0, 0])

    M = se2_exp(-np.array([v, 0, w]) * dt)   # 3x3 SE(2) matrix
    pose = np.array([M[0, 2],
                     M[1, 2],
                     np.arctan2(M[1, 0], M[0, 0])])
    A_k = se2_Ad(pose)

    H = np.array([
    [np.cos(theta), -np.sin(theta), 0],
    [np.sin(theta),  np.cos(theta), 0]
    ])
    Q_body = np.diag(np.array([sigma_v**2, 0, sigma_w**2]))
```

```
    Q_gps = np.diag(np.array([sigma_gps**2, sigma_gps**2]))

    # Propagate state
    X = X @ se2_exp(xi * dt)

    # Prediction
    P = A_k @ P @ A_k.T + Q_body*dt**2

    if not gps_flag:
        return X, P, np.nan

    # Update
    p = np.array([X[0, 2], X[1, 2]])
    S = H @ P @ H.T + Q_gps
    K = P @ H.T @ np.linalg.inv(S)
    P = (np.eye(3) - K @ H) @ P @ (np.eye(3) - K @ H).T + K @ Q_gps @ K.T
    y = z_gps - p
    nu = y @ np.linalg.inv(S) @ y
    de = K @ y

    X = X @ se2_exp(de)

    return X, P, nu
```

In [19]:
```
class Model():
    def __init__(self):
        self.x = sympy.symbols('x')
        self.y = sympy.symbols('y')
        self.theta = sympy.symbols('theta')
        self.dt = sympy.symbols('dt')
        self.v_nom = sympy.symbols('v_nom')
        self.w_ref = sympy.symbols('w_ref')
        self.P = sympy.MatrixSymbol('P', 3, 3)

    def dynamics(self):
        x_next = self.x + self.v_nom * sympy.cos(self.theta)*self.dt
        y_next = self.y + self.v_nom * sympy.sin(self.theta)*self.dt
        theta_next = self.theta + self.w_ref * self.dt
        return sympy.Matrix([x_next, y_next, theta_next])

    def state(self):
        return sympy.Matrix([self.x, self.y, self.theta])

    def Jacobian(self):
        return self.dynamics().jacobian(self.state())

    def Q_body(self):
        sigma_v, sigma_w = sympy.symbols('sigma_v sigma_w')
        return sympy.diag(sigma_v**2, 0, sigma_w**2)

    def Q(self):
        Lk = self.dt * sympy.Matrix([
            [sympy.cos(self.theta), 0, 0],
            [sympy.sin(self.theta), 0, 0],
            [0,                     0, 1]
        ])
```

```
        return Lk * self.Q_body() * Lk.T

    def measurement_noise(self):
        sigma_gps = sympy.symbols('sigma_gps')
        R = sigma_gps**2 * sympy.eye(2)
        return R

    def measurement_jacobian(self):
        return sympy.Matrix([self.x, self.y]).jacobian(self.state())
```

In [20]:
```
class EKF():
    def __init__(self, model):
        syms = [model.x, model.y, model.theta, model.v_nom, model.w_ref, model.dt]
        noise_syms = sympy.symbols('sigma_v sigma_w sigma_gps')

        self.f = sympy.lambdify(syms, model.dynamics(), 'numpy')
        self.F = sympy.lambdify(syms, model.Jacobian(), 'numpy')
        self.Q = sympy.lambdify([model.theta, model.dt, noise_syms[0], noise_syms[1
        self.H = np.array(model.measurement_jacobian()).astype(float)
        self.R = sympy.lambdify([noise_syms[2]], model.measurement_noise(), 'numpy'

    def predict(self, x_est, P_est, v, w, dt, sigma_v, sigma_w):
        x_pred = np.array(self.f(*x_est, v, w, dt)).flatten()
        F_val = np.array(self.F(*x_est, v, w, dt)).astype(float)
        Q_val = np.array(self.Q(x_est[2], dt, sigma_v, sigma_w)).astype(float)
        P_pred = F_val @ P_est @ F_val.T + Q_val
        return x_pred, P_pred

    def update(self, x_pred, P_pred, z, sigma_gps):
        H = self.H
        R = np.array(self.R(sigma_gps)).astype(float)
        K = P_pred @ H.T @ np.linalg.inv(H @ P_pred @ H.T + R)
        x_est = x_pred + K @ (z - H @ x_pred)
        P_est = (np.eye(3) - K @ H) @ P_pred @ (np.eye(3) - K @ H).T + K @ R @ K.T
        return x_est, P_est

    def simulate_step(self, x_est, P_est, z, v, w, dt, sigma_v, sigma_w, sigma_gps)
        x_pred, P_pred = self.predict(x_est, P_est, v, w, dt, sigma_v, sigma_w)
        x_est, P_est = self.update(x_pred, P_pred, z, sigma_gps)
        return x_est, P_est
```

In [21]:
```
# Model
model = Model()
model.dynamics()
kf = EKF(model)
rng = np.random.default_rng(67676767)
```

In [22]:
```
# Constants
pi = np.pi

# Parameters
v_nom = 2
w_turn = 0.4
r_fig = 5
```

```
sigma_v = 1.0
sigma_w = 0.05
sigma_gps = 0.3
dt = 0.1

# Simulation parameters
T_lobe = 2 * pi / w_turn
T_fig8 = 10 * pi
T = 35
n_steps = int(T / dt)
gps_freq = 1

# Initial conditions
x0_true = np.array([0.0, 0.0, 0.0])
x0_est = np.array([0.0, 0.0, 0.0])
P0 = np.diag([0.01, 0.01, 0.01])
H = kf.H
R = np.array(kf.R(sigma_gps)).astype(float)

# Pre-compute reference inputs and GPS schedule
w_ref_vec = np.zeros(n_steps)
gps_flags = np.zeros(n_steps, dtype=bool)
for i in range(n_steps):
    t = i * dt
    w_ref_vec[i] = w_turn if t % T_fig8 < T_lobe else -w_turn
    gps_flags[i] = t % (1 / gps_freq) < 0.01

# Pre-generate noise so all filters use the same realization
noise_v = sigma_v * rng.standard_normal(n_steps)
noise_w = sigma_w * rng.standard_normal(n_steps)
noise_gps = rng.multivariate_normal(np.zeros(2), R, size=n_steps)

# Generate true trajectory and GPS measurements (shared by all filters)
x_true_hist = np.zeros((n_steps + 1, 3))
x_true_hist[0] = x0_true
z_hist = np.full((n_steps, 2), np.nan)

X_true = np.eye(3)
for i in range(n_steps):
    xi = np.array([v_nom + noise_v[i], 0, w_ref_vec[i] + noise_w[i]]) * dt
    X_true = X_true @ se2_exp(xi)
    x_true_hist[i+1] = [X_true[0, 2], X_true[1, 2], np.arctan2(X_true[1, 0], X_true
    if gps_flags[i]:
        z_hist[i] = H @ x_true_hist[i+1] + noise_gps[i]
```

In [23]:

```
x_est_hist = np.zeros((n_steps + 1, 3))
P_est_hist = np.zeros((n_steps + 1, 3, 3))
x_est_hist[0] = x0_est
P_est_hist[0] = P0

x_est = x0_est.copy()
P_est = P0.copy()
for i in range(n_steps):
    x_est, P_est = kf.predict(x_est, P_est, v_nom, w_ref_vec[i], dt, sigma_v, sigma
    if gps_flags[i]:
        x_est, P_est = kf.update(x_est, P_est, z_hist[i], sigma_gps)
```

```
    x_est_hist[i + 1] = x_est
    P_est_hist[i + 1] = P_est
```

In [24]:
```
x_est_hist_es = np.zeros((n_steps + 1, 3))
P_est_hist_es = np.zeros((n_steps + 1, 3, 3))
x_est_hist_es[0] = x0_est
P_est_hist_es[0] = P0

fns = ES_EKF()
es_ekf_fn = fns["error_state_filter"]

x_est = x0_est.copy()
P_est = P0.copy()
for i in range(n_steps):
    gps_flag = 1 if gps_flags[i] else 0
    z_gps = z_hist[i] if gps_flags[i] else np.zeros(2)
    x_est, P_est, _ = es_ekf_fn(
        x_est, P_est, v_nom, w_ref_vec[i], dt,
        sigma_v, sigma_w, sigma_gps, z_gps, gps_flag)
    x_est_hist_es[i + 1] = np.array(x_est).flatten()
    P_est_hist_es[i + 1] = np.array(P_est)
```

In [25]:
```
x_est_hist_iekf = np.zeros((n_steps + 1, 3))
P_est_hist_iekf = np.zeros((n_steps + 1, 3, 3))
x_est_hist_iekf[0] = x0_est
P_est_hist_iekf[0] = P0

X_est = np.eye(3)
P_est = P0.copy()
for i in range(n_steps):
    gps_flag = 1 if gps_flags[i] else 0
    z_gps = z_hist[i] if gps_flags[i] else np.zeros(2)
    X_est, P_est, _ = IEKF(
        X_est, P_est, v_nom, w_ref_vec[i], dt,
        sigma_v, sigma_w, sigma_gps, z_gps, gps_flag)
    x_est_hist_iekf[i + 1] = [X_est[0, 2], X_est[1, 2], np.arctan2(X_est[1, 0], X_e
    P_est_hist_iekf[i + 1] = P_est
```

In [26]:
```
def cov_ellipse(P_2x2, center, n_std=3.0, **kwargs):
    eigvals, eigvecs = np.linalg.eigh(P_2x2)
    order = eigvals.argsort()[::-1]
    eigvals = eigvals[order]
    eigvecs = eigvecs[:, order]
    angle = np.degrees(np.arctan2(eigvecs[1, 0], eigvecs[0, 0]))
    w, h = 2 * n_std * np.sqrt(eigvals)
    return Ellipse(xy=center, width=w, height=h, angle=angle,
                   fill=False, linewidth=1.5, **kwargs)

def body_to_world_cov(P, theta):
    R_th = np.array([[np.cos(theta), -np.sin(theta), 0],
                     [np.sin(theta),  np.cos(theta), 0],
                     [0, 0, 1]])
    return R_th @ P @ R_th.T

fig, ax = plt.subplots(figsize=(8, 8))
```

```
ax.set_aspect('equal')
ax.grid(True)
ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_title('Filter Comparison')

true_line, = ax.plot([], [], 'k-', linewidth=0.8, label='True')
ekf_line, = ax.plot([], [], 'b--', linewidth=1.2, label='EKF')
es_line, = ax.plot([], [], 'r-.', linewidth=1.2, label='ES-EKF')
iekf_line, = ax.plot([], [], color='purple', linestyle=':', linewidth=1.2, label='I
gps_scatter = ax.scatter([], [], c='green', s=15, marker='x', label='GPS', zorder=5
ellipse_patch_ekf = None
ellipse_patch_es = None
ellipse_patch_iekf = None
ax.legend(loc='upper left')

pad = 2
ax.set_xlim(x_true_hist[:, 0].min() - pad, x_true_hist[:, 0].max() + pad)
ax.set_ylim(x_true_hist[:, 1].min() - pad, x_true_hist[:, 1].max() + pad)

skip = 5

def init():
    true_line.set_data([], [])
    ekf_line.set_data([], [])
    es_line.set_data([], [])
    iekf_line.set_data([], [])
    gps_scatter.set_offsets(np.empty((0, 2)))
    return true_line, ekf_line, es_line, iekf_line, gps_scatter

def animate(frame):
    global ellipse_patch_ekf, ellipse_patch_es, ellipse_patch_iekf
    k = frame * skip

    true_line.set_data(x_true_hist[:k+1, 0], x_true_hist[:k+1, 1])
    ekf_line.set_data(x_est_hist[:k+1, 0], x_est_hist[:k+1, 1])
    es_line.set_data(x_est_hist_es[:k+1, 0], x_est_hist_es[:k+1, 1])
    iekf_line.set_data(x_est_hist_iekf[:k+1, 0], x_est_hist_iekf[:k+1, 1])

    z_slice = z_hist[:k]
    gps_mask = ~np.isnan(z_slice[:, 0])
    if gps_mask.any():
        gps_scatter.set_offsets(z_slice[gps_mask])

    if ellipse_patch_ekf is not None:
        ellipse_patch_ekf.remove()
    ellipse_patch_ekf = cov_ellipse(P_est_hist[k, :2, :2], x_est_hist[k, :2], edgec
    ax.add_patch(ellipse_patch_ekf)

    if ellipse_patch_es is not None:
        ellipse_patch_es.remove()
    P_world_es = body_to_world_cov(P_est_hist_es[k], x_est_hist_es[k, 2])
    ellipse_patch_es = cov_ellipse(P_world_es[:2, :2], x_est_hist_es[k, :2], edgeco
    ax.add_patch(ellipse_patch_es)

    if ellipse_patch_iekf is not None:
```

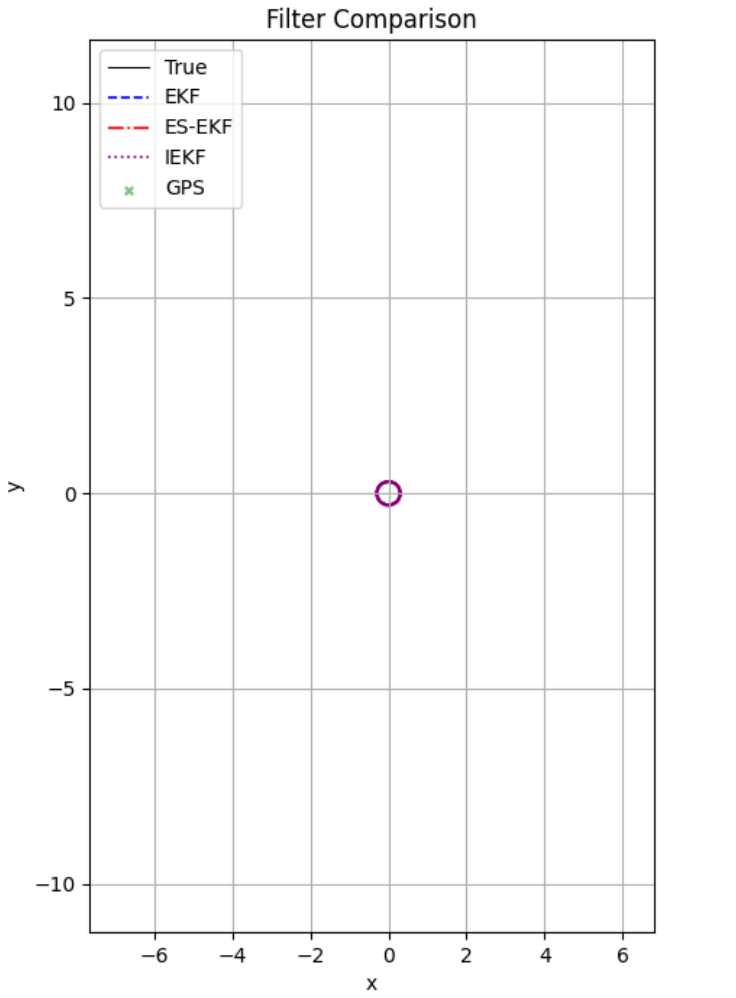
```
        ellipse_patch_iekf.remove()
    P_world_iekf = body_to_world_cov(P_est_hist_iekf[k], x_est_hist_iekf[k, 2])
    ellipse_patch_iekf = cov_ellipse(P_world_iekf[:2, :2], x_est_hist_iekf[k, :2],
    ax.add_patch(ellipse_patch_iekf)

    return true_line, ekf_line, es_line, iekf_line, gps_scatter, ellipse_patch_ekf,

n_frames = (n_steps + 1) // skip
anim = FuncAnimation(fig, animate, init_func=init, frames=n_frames, interval=30, bl
anim.save('filter_comparison.gif', writer='pillow', fps=30)
plt.close(fig)
print('Saved filter_comparison.gif')
HTML(anim.to_jshtml())
```

```
Saved filter_comparison.gif
```

Out[26]:



Once Loop Reflect

In [27]:
```
N = 100
mc_rng = np.random.default_rng(12345)

epsilon_all_ekf  = np.zeros((N, n_steps))
nu_all_ekf       = np.full((N, n_steps), np.nan)
epsilon_all_es   = np.zeros((N, n_steps))
nu_all_es        = np.full((N, n_steps), np.nan)
epsilon_all_iekf = np.zeros((N, n_steps))
nu_all_iekf      = np.full((N, n_steps), np.nan)

fns = ES_EKF()
es_ekf_fn = fns["error_state_filter"]

for j in range(N):
    mc_nv  = sigma_v * mc_rng.standard_normal(n_steps)
    mc_nw  = sigma_w * mc_rng.standard_normal(n_steps)
    mc_gps = mc_rng.multivariate_normal(np.zeros(2), R, size=n_steps)

    X_true_mc = np.eye(3)
    x_true_mc = np.zeros((n_steps + 1, 3))
    x_true_mc[0] = x0_true
    z_mc = np.full((n_steps, 2), np.nan)

    for i in range(n_steps):
        xi = np.array([v_nom + mc_nv[i], 0, w_ref_vec[i] + mc_nw[i]]) * dt
        X_true_mc = X_true_mc @ se2_exp(xi)
        x_true_mc[i+1] = [X_true_mc[0,2], X_true_mc[1,2], np.arctan2(X_true_mc[1,0]
        if gps_flags[i]:
            z_mc[i] = H @ x_true_mc[i+1] + mc_gps[i]

    # --- EKF ---
    x_est = x0_est.copy(); P_est = P0.copy()
    for i in range(n_steps):
        x_true = x_true_mc[i+1]
        x_est, P_est = kf.predict(x_est, P_est, v_nom, w_ref_vec[i], dt, sigma_v, s
        if gps_flags[i]:
            S = H @ P_est @ H.T + R
            innov = z_mc[i] - H @ x_est
            nu_all_ekf[j, i] = innov @ np.linalg.inv(S) @ innov
            x_est, P_est = kf.update(x_est, P_est, z_mc[i], sigma_gps)
        delta_x = x_true - x_est
        delta_x[2] = (delta_x[2] + np.pi) % (2*np.pi) - np.pi
        epsilon_all_ekf[j, i] = delta_x @ np.linalg.inv(P_est) @ delta_x

    # --- ES-EKF ---
    x_est = x0_est.copy(); P_est = P0.copy()
    for i in range(n_steps):
        x_true = x_true_mc[i+1]
        gps_flag = 1 if gps_flags[i] else 0
        z_gps = z_mc[i] if gps_flags[i] else np.zeros(2)
        x_est, P_est, nu_all_es[j, i] = es_ekf_fn(
            x_est, P_est, v_nom, w_ref_vec[i], dt,
            sigma_v, sigma_w, sigma_gps, z_gps, gps_flag)
        x_est = np.array(x_est).flatten()
        P_est = np.array(P_est)
        th = x_est[2]
```

```
        R_rot = np.array([[np.cos(th), np.sin(th)], [-np.sin(th), np.cos(th)]])
        dp_body = R_rot @ (x_true[:2] - x_est[:2])
        dtheta = (x_true[2] - x_est[2] + np.pi) % (2*np.pi) - np.pi
        eps = np.array([dp_body[0], dp_body[1], dtheta])
        epsilon_all_es[j, i] = eps @ np.linalg.inv(P_est) @ eps

    # --- IEKF ---
    X_est = np.eye(3); P_est = P0.copy()
    for i in range(n_steps):
        x_true = x_true_mc[i+1]
        gps_flag = 1 if gps_flags[i] else 0
        z_gps = z_mc[i] if gps_flags[i] else np.zeros(2)
        X_est, P_est, nis = IEKF(
            X_est, P_est, v_nom, w_ref_vec[i], dt,
            sigma_v, sigma_w, sigma_gps, z_gps, gps_flag)
        nu_all_iekf[j, i] = nis
        x_est = np.array([X_est[0,2], X_est[1,2], np.arctan2(X_est[1,0], X_est[0,0]
        th = x_est[2]
        R_rot = np.array([[np.cos(th), np.sin(th)], [-np.sin(th), np.cos(th)]])
        dp_body = R_rot @ (x_true[:2] - x_est[:2])
        dtheta = (x_true[2] - x_est[2] + np.pi) % (2*np.pi) - np.pi
        eps = np.array([dp_body[0], dp_body[1], dtheta])
        epsilon_all_iekf[j, i] = eps @ np.linalg.inv(P_est) @ eps

epsilon_bar      = np.mean(epsilon_all_ekf, axis=0)
nu_bar           = np.nanmean(nu_all_ekf, axis=0)
epsilon_bar_es   = np.mean(epsilon_all_es, axis=0)
nu_bar_es        = np.nanmean(nu_all_es, axis=0)
epsilon_bar_iekf = np.mean(epsilon_all_iekf, axis=0)
nu_bar_iekf      = np.nanmean(nu_all_iekf, axis=0)
```

```
C:\Users\Yahor\AppData\Local\Temp\ipykernel_45668\2578342731.py:82: RuntimeWarning:
Mean of empty slice
  nu_bar           = np.nanmean(nu_all_ekf, axis=0)
C:\Users\Yahor\AppData\Local\Temp\ipykernel_45668\2578342731.py:84: RuntimeWarning:
Mean of empty slice
  nu_bar_es        = np.nanmean(nu_all_es, axis=0)
C:\Users\Yahor\AppData\Local\Temp\ipykernel_45668\2578342731.py:86: RuntimeWarning:
Mean of empty slice
  nu_bar_iekf      = np.nanmean(nu_all_iekf, axis=0)
```

In [28]:
```
nx = 3
nz = 2

# NEES band
nees_lo = chi2.ppf(0.025, df=N*nx) / N
nees_hi = chi2.ppf(0.975, df=N*nx) / N

# NIS band
nis_lo = chi2.ppf(0.025, df=N*nz) / N
nis_hi = chi2.ppf(0.975, df=N*nz) / N
```

In [29]:
```
t_vec = np.arange(1, n_steps + 1) * dt

gps_mask = ~np.isnan(nu_bar)
```

```
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(14, 9), sharex=True)

ax1.plot(t_vec, epsilon_bar, color='b', linewidth=1.5, label=r'$\bar{\epsilon}_k$ E
ax1.plot(t_vec, epsilon_bar_es, color='r', linewidth=1.5, label=r'$\bar{\epsilon}_k
ax1.plot(t_vec, epsilon_bar_iekf, color='purple', linewidth=1.5, label=r'$\bar{\eps
ax1.axhline(nees_lo, color='k', linestyle='--', label='95% bounds')
ax1.axhline(nees_hi, color='k', linestyle='--')
ax1.axhline(nx, color='g', linestyle=':', alpha=0.5, label=r'$n_x = 3$')
ax1.set_ylabel('NEES')
ax1.set_title(f'Time-Averaged NEES ({N} Monte Carlo runs)')
ax1.legend()
ax1.grid(True, alpha=0.3)

ax2.plot(t_vec[gps_mask], nu_bar[gps_mask], color='b', linewidth=1.5, label=r'$\bar
ax2.plot(t_vec[gps_mask], nu_bar_es[gps_mask], color='r', linewidth=1.5, label=r'$\
ax2.plot(t_vec[gps_mask], nu_bar_iekf[gps_mask], color='purple', linewidth=1.5, lab
ax2.axhline(nis_lo, color='k', linestyle='--', label='95% bounds')
ax2.axhline(nis_hi, color='k', linestyle='--')
ax2.axhline(nz, color='g', linestyle=':', alpha=0.5, label=r'$n_z = 2$')
ax2.set_ylabel('NIS')
ax2.set_xlabel('Time [s]')
ax2.set_title(f'Time-Averaged NIS ({N} Monte Carlo runs)')
ax2.legend()
ax2.grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig('NEES_NIS_comparison.png', dpi=150)
plt.show()
```

In [30]:
```
nees_avg_ekf  = np.mean(epsilon_bar)
nees_avg_es   = np.mean(epsilon_bar_es)
nees_avg_iekf = np.mean(epsilon_bar_iekf)
```

```
nis_avg_ekf  = np.nanmean(nu_bar)
nis_avg_es   = np.nanmean(nu_bar_es)
nis_avg_iekf = np.nanmean(nu_bar_iekf)

print(f"{'Filter':<12} {'NEES':>10} {'NIS':>10}")
print("-" * 34)
print(f"{'EKF':<12} {nees_avg_ekf:>10.4f} {nis_avg_ekf:>10.4f}")
print(f"{'ES-EKF':<12} {nees_avg_es:>10.4f} {nis_avg_es:>10.4f}")
print(f"{'IEKF':<12} {nees_avg_iekf:>10.4f} {nis_avg_iekf:>10.4f}")
```

```
Filter             NEES        NIS
----------------------------------
EKF              3.1928     1.9704
ES-EKF           3.1453     1.9563
IEKF             2.9028     1.9533
```